

The Moments of Gila: Closed Forms for Every Panel-Pair Integral of r^m

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Abstract

The companion document *The Integrals of Gila* proved that every moment $I_m(F, F') = \int_F \int_{F'} |\mathbf{x} - \mathbf{y}|^m dS' dS$ of a pair of axis-aligned rectangles is an elementary function of the edge lengths, and left the formulas themselves to be found. This document finds them. Every face pair of two touching cuboid cells on a regular lattice falls into ten Table-1 codes, which resolve into fifteen inequivalent sub-geometries once the offsets of the convolution weights are distinguished; for all fifteen, and for every integer $m \geq -1$, we give closed forms and verified recurrences with explicit elementary seeds, together with a floating-point arrangement whose **Float64** evaluation loses at most 0.91 decimal digits against a 320-bit reference over aspect ratios from 10^{-6} to 10^6 and $m = -1, \dots, 12$. Correctness is at the reference evaluator's own convergence floor, 10^{-44} to 10^{-68} relative, at eleven cell shapes and at all 144 face pairs of the four touching-cell offsets. The series built from these moments reproduces Gila's **wekS**, **wekE**, **wekV** at **intOrd** 48 and 64 and its order-9 fixed rule for the non-touching pairs to the accuracy of those numerics, and exposes a defect in the order-9 rule on slender cells. Four independent derivation families — polar coordinates with the secant-power reduction, the divergence recursion on the product box, the box-potential route, and numerical discovery with exact coefficient recovery — were run without knowledge of each other and converged on the same five primitive blocks; that convergence, and not any single derivation, is the reason to believe the result. The deliverable is `notes/moments/moments.jl`, a standalone Julia file generic in the number type, with `verify.jl` and `gilacmp.jl` producing every table quoted here.

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1 Overview

1.1 What was asked

The task statement in `notes/PROMPT_integrals.md` was: for each of the ten rectangle-pair geometries that occur between touching cells on a regular grid, produce elementary closed forms for I_{-1} and for every odd moment I_{2m+1} , either as an explicit formula valid for general m or as a verified recurrence in m with explicit elementary seeds, together with a cancellation-free floating-point arrangement of each; verify each against an independent high-precision reference to at least 10^{-30} relative for $m = -1, \dots, 12$ at cubic cells, at slender cells with aspect ratios 10^{-3} and 10^3 in each direction and at the two cell shapes Gila actually uses; measure a digits-lost table over aspect ratios 10^{-6} to 10^6 and keep the loss under one decimal digit; demonstrate that the resulting power series reproduces Gila's `wekS`, `wekE`, `wekV` and its order-9 fixed-rule values; and ship a standalone Julia implementation generic over the number type.

The ten codes, in the notation of Table 1 of *The Integrals of Gila*, are S (coincident faces), Ef (coplanar, sharing an edge), Ec (perpendicular, sharing an edge), Vc (coplanar, sharing only a corner), Vp (perpendicular, sharing only a corner), P1 and P2 (parallel, one or two cell lengths apart, aligned), P1s and P2s (the same, laterally shifted by one cell length), and Xg (perpendicular planes whose faces do not meet). Of these, S was already solved for every m ; I_{-1} was known for Ef and Ec; nothing was known for the rest.

1.2 The result

All ten Table-1 codes are done, and they are more than ten geometries. Classifying the 144 face pairs (Δ, F, F') of the four touching-cell offsets by the exact per-axis convolution signature (Section 2.4) splits Xg into four inequivalent sub-geometries, and P1s and P2s into two each, giving **fifteen** sub-codes: S, Ef, Ec, Vc, Vp, P1, P2, P1s, P1s-b, P2s, P2s-b, Xg, Xg-b, Xg-c, Xg-d, with the counts 7, 6, 34, 5, 22, 8, 1, 8, 8, 2, 3, 12, 20, 2, 6 summing to 144. For all fifteen and for every integer $m \geq -1$ there is now a closed form: an explicit elementary expression where one is useful, and always a finite ladder of exact two-term recurrences in m whose seeds at $m = -1$ and $m = 0$ are the elementary functions `asinh`, `atan` and square roots of sums of squares. Against the 320-bit reference evaluator `pairMom` the worst relative error over the fifteen sub-codes, the eleven prescribed cell shapes and $m = -1, \dots, 12$ is 7.35×10^{-44} , with zero of 2310 values above the 10^{-30} bar, and over all 144 face pairs at $m = -1, \dots, 8$ it is 1.91×10^{-48} with zero of 1440 above the bar; in both cases the residual is the reference's own quadrature refinement residual, not the formulas', which order refinement demonstrates by falling 8 to 14 orders per order-step with no floor. The

cancellation-free arrangement loses at most **0.91** decimal digits in **Float64** over the 13×14 aspect grid $a/b, c/b \in [10^{-6}, 10^6]$ and $m = -1, \dots, 12$, with zero cells above one digit out of roughly 37 000 measured. The series assembled from these moments reproduces **wekS**, **wekE**, **wekV** to 6.3×10^{-11} (S), 2.9×10^{-10} (Ef), 9.3×10^{-11} (Ec), 5.0×10^{-10} (Vc) and 1.8×10^{-9} (Vp) at **intOrd** 48, with the order-64 values lying between the order-48 values and the series, and reproduces the order-9 fixed rule for the non-touching pairs to 10^{-16} – 10^{-13} on cubic cells.

1.3 How to use it

The deliverable is **notes/moments/moments.jl**: 1720 lines, no package dependencies, generic in the number type (**BigFloat** in gives **BigFloat** out; **Float32** works; **Rational** panels give exactly the values of the corresponding float panels). A panel is a triple of coordinate intervals with one axis degenerate, $((x1, x2), (y1, y2), (z1, z2))$. The three entry points are

call	returns
pairMoments (A, B, mMax; tol = nothing)	Vector with out[m+2] = I_m , $m = -1, \dots, mMax$
faceMoments (D, F, Fp, s, mMax; tol = nothing)	the same for face F of a cell at offset D against F'
momentSeries (A, B, f, mMax)	(value, rel): $\int_A \int_B g$ and $ last\ term / sum $

Even m is returned as well as odd; it costs nothing, since the same ladder carries both parities. **tol** defaults to **nothing**, which makes the box breakpoints exactly rational; passing a number instead merges breakpoints closer than **tol** relative to the axis extent, which is the right choice when the panel coordinates are not exactly representable (Section 5.6). **momentSeries** sums $\frac{1}{4\pi f^2} \sum_n \frac{(2\pi i f)^n}{n!} I_{n-1}$ and returns its own convergence indicator; that indicator is an honest upper bound on the truncation error in all 240 rows of the audit's series sweep, over-stating by a factor between 3.4 and more than 10^3 and never under-stating. **mMax** = 12 is a $\lambda/8$ tool: at a $\lambda/4$ cell it leaves up to 7.5×10^{-3} relative error on the far sub-codes, and **mMax** = 30 removes that.

The file also carries the plain explicit closed forms of Sections 3 and 4 (**momSlf**, **momFlt**, **momVtxCop**, **momPar**, **prpMom**, **vcMom**, **vpMom**, **xgMom**). They are cross-checks only: they are correct but are *not* arranged against cancellation, and lose up to 22.6 digits in **Float64**. Every actual evaluation should go through **pairMoments**.

Verification is reproduced by

```
JULIA_NUM_THREADS=1 julia --startup-file=no verify.jl a|b|c|d|e
```

which writes **tables/{a,b,c,d,e}*.txt**: (a) correctness at the eleven shapes, (b) all 144 face pairs against **pairMom**, (c) cross-checks against the explicit forms plus homogeneity, symmetries, orientation invariance and the **tol** test, (d) the digits-lost grids, (e) timing. The whole run is 7 minutes 5 seconds single-threaded; only part (b) calls **pairMom** and only part (c) calls **pairKer**. Part (a) reads **refcache.txt**, a 1155-line cache of 320-bit **pairMom** values that took 36.7 minutes to build, so no cached case is recomputed.

The comparison with Gila's own numerics is

```
JULIA_NUM_THREADS=1 julia --startup-file=no --project=<env> gilacmp.jl a|b|c
```

with **<env>** any environment in which using **GilaElectromagnetics** works. It writes **tables/f_gila-{a,b,c}.txt**: (a) **wekS/wekE/wekV** at **intOrd** 48 and 64 against the series, (b) the order-9 fixed rule on every non-touching face pair, (c) the frequency homogeneity of the series terms. Part (a) takes 37 minutes, 36 of them inside Gila's own weak integrals; part (b) 35 minutes, nearly all of it in 280 **BigFloat** series sums; part (c) 5 seconds.

1.4 How to read this document

Section 2 is the part worth reading first: it sets out the single decomposition that all fifteen geometries share, the five primitive blocks it terminates in, and the fact that four independent derivation families found the same blocks. Sections 3 and 4 then treat the coplanar/parallel and the perpendicular families, one subsection per sub-code, each with its figure, its derivation, its boxed formula or recurrence with seeds, the explicit low- m forms where they exist, its cancellation-free arrangement and its measured numbers. Section 5 treats the cancellation-free arrangement as a subject in its own right, since that was where most of the work went and where most of the surprises were. Section 6 collects the verification tables, Section 7 the comparison with Gila. Section 8 is the registry of approach families, written so that the reader learns from what was blocked as well as from what worked. Section 9 is the honest account of what remains uncertain, and Section 10 the implementation notes for wiring the result into Gila.

Conventions follow the companion document. Lengths are in wavelengths, the frequency enters through the single complex scalar f with $k = 2\pi f$, and Gila's scalar kernel is $g(r) = e^{2\pi i f r} / (4\pi f^2 r)$. A cell has edge lengths $\mathbf{s} = (s_1, s_2, s_3)$. Where a formula involves named lengths I write a, b, c for the three independent edge lengths of the canonical representative of a geometry, and every orientation of that geometry in the tables is covered by a permutation of those arguments, which was verified rather than assumed (Section 6).

2 The unifying structure

2.1 Difference variables and the product box

Every geometry reduces the same way. The kernel depends only on the separation $\mathbf{w} = \mathbf{x} - \mathbf{y}$, so for each axis independently the pair of coordinate intervals $[a_1, a_2]$ (target) and $[b_1, b_2]$ (source) contributes the convolution of their indicators: a trapezoid supported on $[b_1 - a_2, b_2 - a_1]$ with slopes ± 1 and plateau at height $\min(a_2 - a_1, b_2 - b_1)$. If one interval is degenerate (a rectangle in a perpendicular plane has zero extent along one axis) the weight is an indicator; if both are degenerate the axis contributes a frozen offset. On the lattice the trapezoid is always a triangle, because both intervals have the same length, and there are exactly three shapes it can take: centred on $[-p, p]$ (aligned intervals), shifted to $[0, 2p]$ (adjacent intervals), and reflected copies of these.

Splitting each trapezoid at its breakpoints and reflecting into the positive octant gives what `pairBxs` returns: a list of axis-aligned boxes with a multilinear weight,

$$I_m = \sum_{\text{boxes}} \int_{lo_1}^{hi_1} \cdots \int_{lo_d}^{hi_d} \left[\prod_{i=1}^d (c_i + d_i u_i) \right] \left(\sum_{i=1}^d u_i^2 + g^2 \right)^{m/2} du_1 \cdots du_d, \quad (1)$$

with $d = 2$ and a frozen offset $g^2 \geq 0$ for the coplanar and parallel pairs (S, Ef, Vc, P1, P2, P1s, P1s-b, P2s, P2s-b), and $d = 3$ with $g = 0$ for the perpendicular pairs (Ec, Vp, Xg, Xg-b, Xg-c, Xg-d). The decomposition is `pairBox` of `notes/gen/ref/mom.jl` unchanged; only the box integrator is replaced. Each per-axis weight is one of exactly three kinds on the lattice,

$$\text{rise } (u - lo) \text{ on } [0, p], \quad \text{fall } (hi - u) \text{ on } [0, p], \quad \text{fall } (2p - u) \text{ on } [p, 2p], \quad (2)$$

plus the indicator 1, and every box carries a positive weight, so the assembly $I_m = \sum_{\text{boxes}} (\cdots)$ introduces no cancellation of its own. Nine box types occur in two dimensions and a comparable number in three; they are enumerated and measured individually in Section 5.

Two facts about (1) organize everything that follows. First, the offset g is the *only* way the third dimension enters a coplanar pair, which is why P1, P2, P1s, P2s, P1s-b and P2s-b are S, Ef and Vc with $g = s_3$ or $g = 2s_3$ and nothing else changed. Second, a perpendicular pair has exactly one axis carrying a trapezoid weight (the shared axis) and two carrying indicators, which is why Ec, Vp, Xg, Xg-b, Xg-c and Xg-d differ only in where those indicators sit and where the triangle is centred.

2.2 The divergence identity and the primitive ladder

The engine is the divergence identity of Theorem 5.1 of the companion document, applied on each box. For Y the position vector in N' dimensions and $\phi_m = (|Y|^2 + c^2)^{m/2}$,

$$\nabla \cdot (Y \phi_m) = (N' + m) \phi_m - m c^2 \phi_{m-2}, \quad (3)$$

so integrating over a box turns an N' -dimensional moment of exponent m into a boundary sum of $(N' - 1)$ -dimensional moments of exponent m plus c^2 times the same N' -dimensional moment at exponent $m - 2$. The recursion lowers dimension and exponent together and terminates in one dimension. The c^2 term is exactly what the parallel offset geometries need, and it is what makes one ladder cover $m = -1, 1, 3, \dots$ at constant cost per step.

Run to the end, (3) produces five blocks and nothing else.

One dimension.

$$J_n(L; d) = \int_0^L (t^2 + d^2)^{n/2} dt, \quad J_{-1} = \operatorname{asinh} \frac{L}{d}, \quad J_0 = L, \quad J_n = \frac{L (L^2 + d^2)^{n/2} + n d^2 J_{n-2}}{n+1}. \quad (4)$$

Unrolled, with $s = \sqrt{L^2 + d^2}$ and $n = 2k - 1$,

$$\begin{aligned} J_{2k-1}(L; d) &= \frac{(2k-1)!!}{(2k)!!} \left[L \sum_{i=1}^k \frac{(2i-2)!!}{(2i-1)!!} s^{2i-1} d^{2(k-i)} + d^{2k} \operatorname{asinh} \frac{L}{d} \right], \\ J_{2k}(L; d) &= \sum_{j=0}^k \binom{k}{j} d^{2(k-j)} \frac{L^{2j+1}}{2j+1}. \end{aligned} \quad (5)$$

The polar route reaches the same object as $J_K(L, g) = g^{K+1} S_{K+2}(L/g)$ with $S_N = \int_0^{\arctan t} \sec^N \theta d\theta$, which is how the double-factorial coefficients appear there.

Two dimensions, the rectangle moment. With $d_A = \sqrt{a^2 + c^2}$, $d_B = \sqrt{b^2 + c^2}$, $R = \sqrt{a^2 + b^2 + c^2}$ and $\phi_m = (u^2 + v^2 + c^2)^{m/2}$,

$$M_{ij}^{(m)}(a, b; c) = \int_0^a \int_0^b u^i v^j \phi_m dv du, \quad i, j \in \{0, 1\}, \quad (6)$$

and (3) with $N' = 2$ gives the ladder and its two seeds,

$$\begin{aligned} (m+2) M_{00}^{(m)}(a, b; c) &= a J_m(b; d_A) + b J_m(a; d_B) + m c^2 M_{00}^{(m-2)}(a, b; c), \\ M_{00}^{(-1)}(a, b; c) &= a \operatorname{asinh} \frac{b}{d_A} + b \operatorname{asinh} \frac{a}{d_B} - c \operatorname{atan} \frac{ab}{cR}, \quad M_{00}^{(0)} = ab. \end{aligned} \quad (7)$$

The $m = -1$ seed is the Newtonian potential of the rectangle $[0, a] \times [0, b]$ at height c above its corner. The three weighted moments do not need a ladder at all: doing the u -integral first is exact, and

$$\boxed{M_{10}^{(m)}(a, b; c) = \frac{J_{m+2}(b; d_A) - J_{m+2}(b; c)}{m+2}, \quad M_{11}^{(m)}(a, b; c) = \frac{R^{m+4} - d_A^{m+4} - d_B^{m+4} + c^{m+4}}{(m+2)(m+4)},} \quad (8)$$

with $M_{01}^{(m)}(a, b; c) = M_{10}^{(m)}(b, a; c)$. The M_{11} closed form was checked against its own divergence recurrence $(m+4)M_{11}^{(m)} = [a^2(R^{m+2} - d_A^{m+2}) + b^2(R^{m+2} - d_B^{m+2})]/(m+2) + mc^2 M_{11}^{(m-2)}$ to 3.6×10^{-81} , rather than assumed.

Three dimensions, the box moment. With $r^2 = u^2 + v^2 + w^2$ on the corner box $[0, a] \times [0, b] \times [0, c]$, (3) with $N' = 3$, $c = 0$ gives

$$\boxed{\begin{aligned} (m+3) K^{(m)}(a, b, c) &= a M_{00}^{(m)}(b, c; a) + b M_{00}^{(m)}(a, c; b) + c M_{00}^{(m)}(a, b; c), \\ K_u^{(m)}(a, b, c) &= \int u r^m = \frac{M_{00}^{(m+2)}(b, c; a) - M_{00}^{(m+2)}(b, c; 0)}{m+2}, \end{aligned}} \quad (9)$$

and the same offset-difference construction gives $\int uv r^m$ and $\int uvw r^m$ from M_{10} and M_{11} . The identity $K^{(m)}$ needs no separate seed, because $M_{00}^{(m)}$ already has one.

The check that ties it together. At $m = -1$ the K recursion collapses *algebraically* onto the classical box potential: substituting the $M_{00}^{(-1)}$ seed three times, each asinh appears twice and each atan once, giving

$$\begin{aligned} K^{(-1)}(a, b, c) = N_0(a, b, c) &= ab \operatorname{asinh} \frac{c}{h_{ab}} + bc \operatorname{asinh} \frac{a}{h_{bc}} + ca \operatorname{asinh} \frac{b}{h_{ca}} \\ &\quad - \frac{a^2}{2} \operatorname{atan} \frac{bc}{aR} - \frac{b^2}{2} \operatorname{atan} \frac{ca}{bR} - \frac{c^2}{2} \operatorname{atan} \frac{ab}{cR}, \end{aligned} \quad (10)$$

equation (22) of the companion document. This is an algebraic identity, so the K recursion and the M_{00} seed corroborate each other independently of any quadrature. Checked to 1.79×10^{-96} .

The weighted shared-axis primitive. A perpendicular pair carries a triangle weight on the shared axis, and the object it needs is

$$P_m(T; B, C) = \int_0^T (T-x) \Psi_m(x; B, C) dx, \quad \Psi_m(t; B, C) = M_{00}^{(m)}(B, C; t), \quad (11)$$

that is, the rectangle moment Ψ integrated against the triangle weight along the perpendicular direction. (Ψ and M_{00} are the same primitive; the two names come from the two derivation routes and differ only in which argument is called the offset.) The key identity, and the one that removes the only genuinely awkward integral in the whole problem, is

$$\boxed{P_m(T; B, C) = T K^{(m)}(T, B, C) - \frac{\Psi_{m+2}(T; B, C) - \Psi_{m+2}(0; B, C)}{m+2}.} \quad (12)$$

It holds because $x r^m dx$ is an exact differential: the x -weighted part of the integral is a plain difference of Ψ at exponent $m+2$, and the unweighted part is the box moment. *No antiderivative*

of an arctangent is ever needed — which is worth saying, because integrating N_0 of (10)-type by parts in the shared direction looks at first as though it must produce one. Equation (12) reproduces the existing closed form `rSrfEdgCrn` (equation (23) of the companion document) term for term at $m = -1$.

2.3 Four routes, one set of blocks

The five blocks J_n , M_{00}/Ψ , M_{10} , M_{11} , K and the identity (12) were not designed and then implemented. Four sub-agents were given four different mathematical mechanisms and were deliberately not told what the others were doing:

- the **polar** route integrated the radial variable exactly on each corner box and reduced the angular integrals by the secant-power recursion $\int \sec^N = \frac{\sec^{N-2} \tan}{N-1} + \frac{N-2}{N-1} \int \sec^{N-2}$, generalizing Section 5.2 of the companion document to the trapezoidal weights;
- the **divergence** route implemented Theorem 5.1’s proof as a symbolic program lowering dimension and exponent together;
- the **box-potential** route reduced each perpendicular pair to $\int (\text{weight}) \Phi_m(u) du$ with Φ_m the potential of a rectangle at a point, and integrated along the shared direction;
- the **discovery** route computed moments to 200 bits at rational geometry, conjectured a basis of elementary atoms, and recovered the rational coefficients by linear algebra before deriving them.

All four terminated in the same blocks. The polar route’s $J_K(L, g) = g^{K+1} S_{K+2}(L/g)$ is (4); its four corner moments G_{ij} are M_{ij} ; its Φ is the $M_{00}^{(-1)}$ seed. The box-potential route’s Ψ , K and P are (7), (9) and (12). The discovery route’s $L^{(m)}$, $P^{(m)}$, $N^{(m)}$, $N_x^{(m)}$, M_{ij} are the same five objects again. Cross-checked numerically, the polar and divergence routes agree to 5.9×10^{-86} and the explicit discovery forms agree with their own recursions to 5.5×10^{-78} (Vc), 5.6×10^{-79} (Vp) and 2.0×10^{-66} (Xg). That three independent derivations of a long computation land on the same five blocks is the strongest structural evidence in this document; the numerical agreement with `pairMom` is the strongest numerical evidence.

2.4 The sub-geometry catalogue

Table 1 of the companion document classifies face pairs by a geometric description (coplanar or perpendicular, touching at a face, an edge or a corner, one or two cells apart, shifted or not). That classification is coarser than the integral. Two pairs give the same integral if and only if their per-axis convolution weights agree, so the right invariant is the *signature*: for each axis, the kind of weight (frozen offset, trapezoid, indicator) and its support, computed exactly on the unit cell and sorted. Under that invariant, 105 of the 144 pairs match one of the ten canonical signatures and 39 do not; the 39 fall into five further classes.

sub-code	count	signature (sorted per-axis)	representative (Δ, F, F')
S	7	(0, 0) (1, -1, 1) (1, -1, 1)	(0, 0, 0) yzL-yzL
Ef	6	(0, 0) (1, -1, 1) (1, 0, 2)	(1, 0, 0) xzL-xzL
Ec	34	(1, -1, 1) (2, 0, 1) (2, 0, 1)	(0, 0, 0) yzL-xzL
Vc	5	(0, 0) (1, 0, 2) (1, 0, 2)	(1, 1, 0) xyL-xyL
Vp	22	(1, 0, 2) (2, 0, 1) (2, 0, 1)	(1, 0, 0) xzL-xyL
P1	8	(0, 1) (1, -1, 1) (1, -1, 1)	(0, 0, 0) yzL-yzU
P2	1	(0, 2) (1, -1, 1) (1, -1, 1)	(1, 0, 0) yzU-yzL
P1s	8	(0, 1) (1, -1, 1) (1, 0, 2)	(1, 0, 0) xzL-xzU
P1s-b	8	(0, 1) (1, 0, 2) (1, 0, 2)	(1, 1, 0) xyL-xyU
P2s	2	(0, 2) (1, -1, 1) (1, 0, 2)	(1, 1, 0) yzU-yzL
P2s-b	3	(0, 2) (1, 0, 2) (1, 0, 2)	(1, 1, 1) yzU-yzL
Xg	12	(1, -1, 1) (2, 0, 1) (2, 1, 2)	(1, 0, 0) yzU-xzL
Xg-b	20	(1, 0, 2) (2, 0, 1) (2, 1, 2)	(1, 1, 0) yzU-xyL
Xg-c	2	(1, -1, 1) (2, 1, 2) (2, 1, 2)	(1, 1, 0) yzU-xzL
Xg-d	6	(1, 0, 2) (2, 1, 2) (2, 1, 2)	(1, 1, 1) yzU-xzL

The signature entries are (0, |offset|) for a frozen axis, (1, *lo*, *hi*) for a trapezoid and (2, *lo*, *hi*) for an indicator, with the support reflected to non-negative values. The five new classes differ from their Table-1 parents in exactly two ways: whether the triangle on the shared axis is centred ($\delta = 0$) or shifted to touch ($\delta = 1$), and whether the indicator axes sit at (gap 0, gap 1) or at (gap 1, gap 1):

- **Xg-b**: trapezoid axis shifted, indicators (gap 1, gap 0);
- **Xg-c**: trapezoid axis centred as in Xg, but *both* indicator axes a full cell away;
- **Xg-d**: trapezoid axis shifted *and* both indicators a full cell away;
- **P1s-b**, **P2s-b**: both in-plane axes shifted, where the canonical P1s and P2s have one shifted and one fully overlapping.

The classification was verified numerically, not merely combinatorially: each of the 144 face pairs was evaluated against its canonical representative under the matching axis permutation at $s = (3/7, 5/11, 2/3)$ in 256-bit arithmetic, giving 144/144 agreement with a maximum relative error of 1.60×10^{-74} . The full map is Appendix A.

This split matters, and it is the reason a document about “ten geometries” has fifteen sections. Two of the five new classes, P1s-b and P2s-b, have no explicit closed form in any of the four derivation families: they were found by the signature classification and are covered only by the box route. The Xg family’s four members have visibly different formulas, and the discovery route’s independent finding that the Xg variant with a shifted triangle and one gap-1 indicator has *exactly* Vp’s reduced integral is the same observation arrived at from the other side.

3 The coplanar and parallel-offset geometries

Nine of the fifteen sub-codes are two-dimensional in the sense of (1): both panels are parallel to a common plane, the two in-plane axes carry trapezoid weights, and the third axis contributes a frozen offset $c \in \{0, s_3, 2s_3\}$. They are S, Ef and Vc at $c = 0$; P1, P1s and P1s-b at $c = s_3$; P2, P2s and P2s-b at $c = 2s_3$. The three families differ only in which trapezoids appear.

3.1 The common assembly

Write the weight of a box as $(\alpha_0 + \alpha_1 u)(\beta_0 + \beta_1 v)$ and let

$$C(P, Q) = \alpha_0 \beta_0 M_{00}^{(m)}(P, Q; c) + \alpha_0 \beta_1 M_{01}^{(m)}(P, Q; c) + \alpha_1 \beta_0 M_{10}^{(m)}(P, Q; c) + \alpha_1 \beta_1 M_{11}^{(m)}(P, Q; c), \quad (13)$$

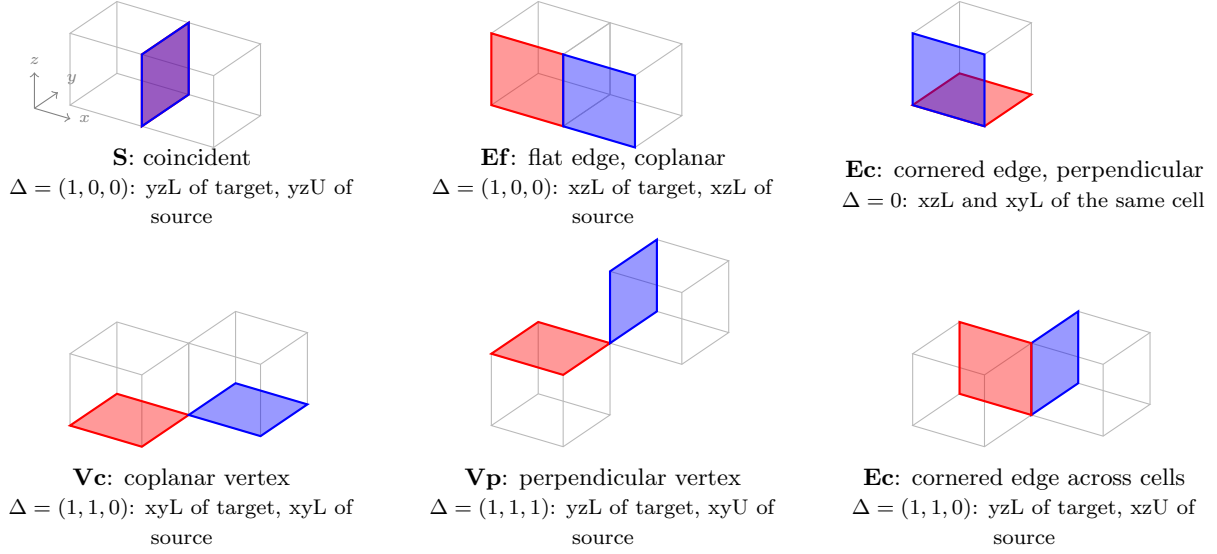


Figure 1: The five contact geometries of Table 1. Red: source face F' (source cell at the origin); blue: target face F (target cell displaced by Δ). Where the two faces coincide (S) the colours overlap.

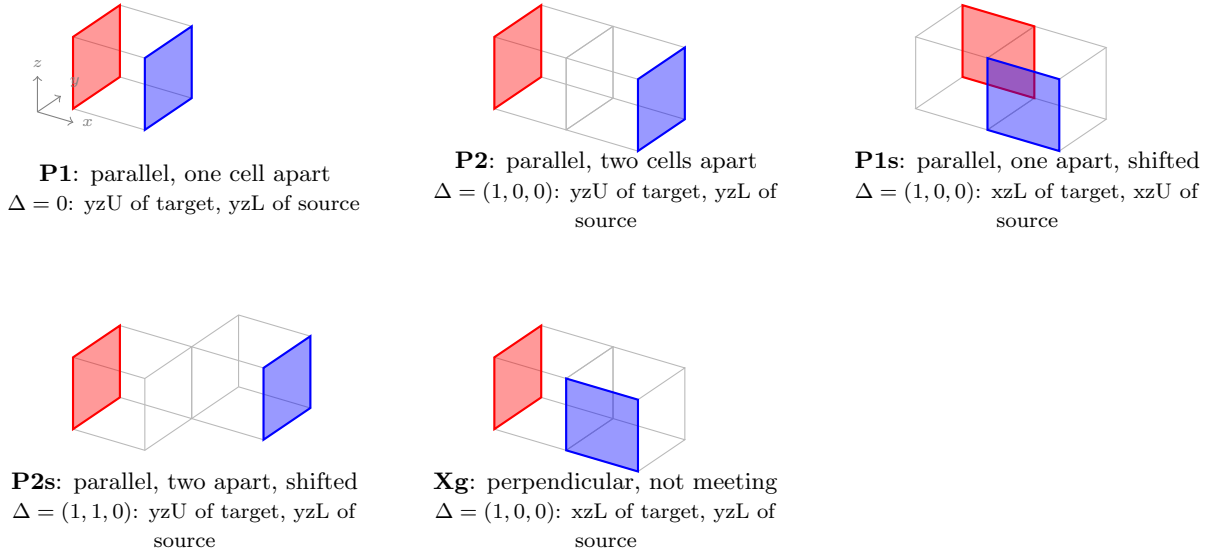


Figure 2: The five non-touching face-pair geometries that occur between touching cells. Same colour convention.

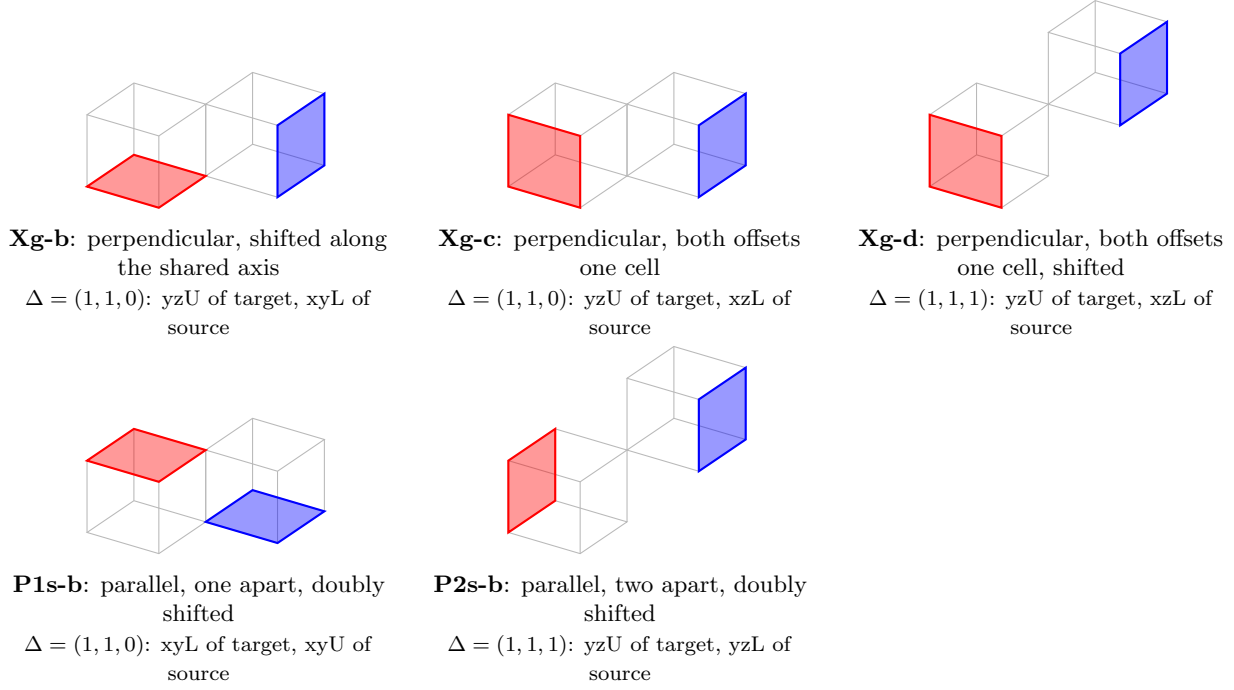


Figure 3: The five sub-geometries that the signature classification separates from their Table-1 parents: three further Xg classes, and the doubly shifted parallel classes. Same colour convention.

$$\mathcal{R}[u_1, u_2; v_1, v_2; \alpha, \beta] = C(u_2, v_2) - C(u_1, v_2) - C(u_2, v_1) + C(u_1, v_1), \quad (14)$$

the corner inclusion–exclusion that turns corner moments into a moment over $[u_1, u_2] \times [v_1, v_2]$. Then, with p, q the two in-plane edge lengths, w the width across a shared edge, e the shared edge length, and a, b the two in-plane edges of a corner-touching pair,

$$\begin{aligned} \text{S, P1, P2 } (c = 0, s_3, 2s_3) : \quad I_m &= 4 \mathcal{R}[0, p; 0, q; (p, -1), (q, -1)], \\ \text{Ef, P1s, P2s } (c = 0, s_3, 2s_3) : \quad I_m &= 2 \{ \mathcal{R}[0, w; 0, e; (0, 1), (e, -1)] \\ &\quad + \mathcal{R}[w, 2w; 0, e; (2w, -1), (e, -1)] \}, \\ \text{Vc, P1s-b, P2s-b } (c = 0, s_3, 2s_3) : \quad I_m &= \mathcal{R}[0, a; 0, b; (0, 1), (0, 1)] \\ &\quad + \mathcal{R}[a, 2a; 0, b; (2a, -1), (0, 1)] \\ &\quad + \mathcal{R}[0, a; b, 2b; (0, 1), (2b, -1)] \\ &\quad + \mathcal{R}[a, 2a; b, 2b; (2a, -1), (2b, -1)]. \end{aligned} \quad (15)$$

The factor 4 for S and the factor 2 for Ef are the reflections of the difference vector into the positive quadrant; Vc’s four terms are the four quadrants of the shifted triangle pair and carry no reflection factor, because its trapezoids are one-sided.

Two remarks. First, *the three offset values are the only difference between the three rows of Table 1 they cover*: a parallel-offset geometry is its coplanar parent with $c \neq 0$, which enters only through the c^2 term of (3) and the c -powers of M_{00} , so no new derivation is required and none was done. Second, the corner inclusion–exclusion in $\mathcal{R}$ is a genuine source of cancellation for the shifted boxes, and Section 5 replaces it: the shipped code integrates each shifted box directly rather than as a four-corner difference.

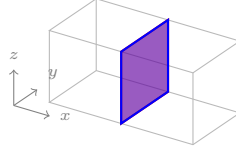
For the explicit forms it is convenient to have the four corner moments in the notation the polar route used, with $\alpha = \sqrt{p^2 + c^2}$, $\beta = \sqrt{q^2 + c^2}$ and $R = \sqrt{p^2 + q^2 + c^2}$:

$$G_{ij}(p, q; M, c) = \int_0^p \int_0^q u^i v^j (u^2 + v^2 + c^2)^{M/2} = M_{ij}^{(M)}(p, q; c), \quad (16)$$

$$G_{00} = \frac{1}{M+2} \left[\sum_{\substack{k \equiv M \pmod{2} \\ k_0 \leq k \leq M}} c^{M-k} (p J_k(q; \alpha) + q J_k(p; \beta)) + [M \text{ odd}] c^{M+1} \Phi(p, q, c) \right], \quad (17)$$

with $k_0 = 1$ for odd M and 0 for even M , and $\Phi(p, q, c) = p \operatorname{asinh}(q/\alpha) + q \operatorname{asinh}(p/\beta) - c \operatorname{atan}(pq/(cR))$ the rectangle potential, which is $M_{00}^{(-1)}$ again.

3.2 S: coincident faces



$\Delta = (1, 0, 0)$: yzL of the target against yzU of the source. Panels $A = B = [0, a] \times [0, b] \times \{0\}$.

This is the case Section 5.2 of the companion document solved for every m at once, and it is used here as the template and as a test of the machinery rather than as new work. The difference-variable reduction gives the bilinear weight $(p - u)(q - v)$ on the quadrant, and in polar coordinates about the corner the radial integral is exact. With $h = \sqrt{p^2 + q^2}$ and $\operatorname{mom}(m) = \int_0^p \int_0^q (p - u)(q - v)(u^2 + v^2)^{m+1/2}$, so that $I_{2m+1} = 4 \operatorname{mom}(m)$,

$$\begin{aligned} \operatorname{mom}(m) = B_m & \left[p^2 q^2 \sum_{j=1}^{m+1} \frac{(2j-2)!!}{(2j-1)!!} h^{2j-1} (p^{2(m+1-j)} + q^{2(m+1-j)}) + p q^{2m+4} \operatorname{asinh} \frac{p}{q} + q p^{2m+4} \operatorname{asinh} \frac{q}{p} \right] \\ & - \frac{h^{2m+5} - p^{2m+5} - q^{2m+5}}{(2m+3)(2m+4)(2m+5)}, \quad B_m = \frac{(2m+1)!!}{(2m+2)!!(2m+3)(2m+4)}, \end{aligned} \quad (18)$$

which is equation (31) of the companion document, and its two-term recurrence, with $X_m(p, q) = q p^{2m+4} S_{2m+3}$,

$$X_{-1} = p^2 q \operatorname{asinh} \frac{q}{p}, \quad X_m = \frac{p^2 q^2 h^{2m+1}}{2m+2} + \frac{2m+1}{2m+2} p^2 X_{m-1}, \quad (19)$$

$$\operatorname{mom}(m) = \frac{X_m(p, q) + X_m(q, p)}{(2m+3)(2m+4)} - \frac{h^{2m+5} - p^{2m+5} - q^{2m+5}}{(2m+3)(2m+4)(2m+5)}. \quad (20)$$

At $m = -1$ the sum is empty and (18) is **rSrfs1f**. The box route reaches the same value from (15) with $c = 0$; the two agree to 2.33×10^{-90} over the eleven shapes and $m = -1, \dots, 12$, which is the cross-check that the general assembly is right where the answer was already known.

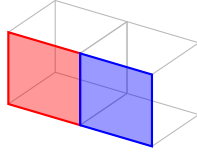
Cancellation. The only dangerous group is $h^{2m+5} - q^{2m+5}$, a difference of two $O(q^{2m+5})$ numbers costing $2 \log_{10}(q/p)$ digits. Written as the product $h^n - b^n = \frac{a^2}{h+b} \sum_{i=0}^{n-1} h^i b^{n-1-i}$ with $(a, b) = \min \max(p, q)$, every remaining term is positive. In the shipped code S never reaches that expression at all: its four boxes are corner boxes of type fall/fall at offset $c = 0$, and the box integrator handles them directly.

Measured. Correctness 6.34×10^{-44} (thinX, $m = -1$; reference floor), 144-pair sweep 1.91×10^{-48} , cross-check against (18) 2.33×10^{-90} . Digits lost in `Float64` over the 13×14 grid, by m :

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
S	0.30	0.00	0.07	0.00	0.00	0.00	0.29	0.00	0.30	0.00	0.30	0.05	0.40	0.39

worst 0.40 at $(a/b, c/b) = (10^3, 10^{-6})$, $m = 11$. Against Gila: the series reproduces `wekS` to 6.30×10^{-11} at `intOrd` 48 and 1.37×10^{-11} at 64 (worst over the cell scales carrying both orders), and I_{-1} agrees with $4\pi f^2$ `rSrfSlf` to 2.51×10^{-16} .

3.3 Ef: coplanar, sharing an edge



$\Delta = (1, 0, 0)$: xzL of the target against xzL of the source. Panels $A = [0, a] \times [0, b] \times \{0\}$, $B = [a, 2a] \times [0, b] \times \{0\}$; shared edge b , width a .

In the direction across the shared edge the convolution weight is the triangle $a - |u - a|$ on $(0, 2a)$, and in the direction along it the triangle $b - |v|$. Splitting at $u = a$ gives the two boxes of the middle row of (15): weight $u(b - v)$ on $[0, a] \times [0, b]$ and weight $(2a - u)(b - v)$ on $[a, 2a] \times [0, b]$. Both are the corner moments M_{ij} , and at $m = -1$ collecting terms gives `rSrfEdgFlt`, equation (17) of the companion document,

$$\boxed{I_{-1}(\text{Ef}) = \frac{1}{3} [Q + 6a^2b D_1 + 6ab^2 D_2]}, \quad \begin{aligned} Q &= 2h_1^3 - h_2^3 - a^3 + 6b^3, \\ D_1 &= \operatorname{asinh} \frac{2b}{a} - \operatorname{asinh} \frac{b}{a}, \\ D_2 &= 2 \operatorname{asinh} \frac{a}{2b} - \operatorname{asinh} \frac{a}{b}, \end{aligned} \quad (21)$$

with $h_1 = \sqrt{a^2 + b^2}$, $h_2 = \sqrt{a^2 + 4b^2}$. (The companion document's `rSrfEdgFlt` carries the kernel's $1/(4\pi f^2)$; here I_{-1} is the bare integral, so the prefactor is $1/3$ rather than $1/(12\pi f^2)$.) For general m there is no reason to collect: the assembly (15) with the general- m G_{ij} of (17)–(8) is a closed form for every integer $m \geq -1$, and it is what the code evaluates.

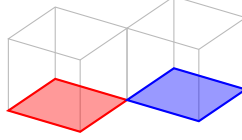
Cancellation. As written the collected $m = -1$ form cancels catastrophically when $a \ll b$: Q is $O(a^3b^2/b^3)$ out of terms of size b^3 , and the two asinh differences are small differences of large logarithms. The remedies are the companion document's: write Q as a manifestly negative product and each difference as $\log(1 + x)$ with x a ratio of positive quantities. The shipped code again does not use the collected form; its two box types (fall₁/fall₀ and rise₀/fall₀) are integrated directly.

Measured. Correctness 3.82×10^{-44} (thinX, $m = -1$), 144-pair sweep 1.44×10^{-48} , cross-check against the polar explicit form 8.10×10^{-90} , and against `rSrfEdgFlt` at $m = -1$, 1.9×10^{-84} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
Ef	0.05	0.00	0.00	0.05	0.24	0.03	0.03	0.14	0.17	0.09	0.21	0.34	0.37	0.26

worst 0.37 at $(10^3, 10^{-6})$, $m = 11$. Against Gila: **wekE** entries 1–6 agree with the series to 2.90×10^{-10} at **intOrd** 48 and 1.92×10^{-11} at 64; I_{-1} against $4\pi f^2$ **rSrfEdgFlt** is 1.02×10^{-16} . Ef is also the one entry where Gila’s order refinement moves the wrong way on the slender cell (Section 7).

3.4 Vc: coplanar, sharing only a corner



$\Delta = (1, 1, 0)$: xyL of the target against xyL of the source. Panels $A = [0, a] \times [0, b] \times \{0\}$, $B = [a, 2a] \times [b, 2b] \times \{0\}$.

Both in-plane weights are shifted triangles, so the four quadrants of (15) all survive and none of them touches the origin except at one corner. Nothing was known about this geometry before; it now has an explicit closed form for every m , obtained independently by the polar route and by numerical discovery with exact coefficient recovery. With $h_{ab} = \sqrt{a^2 + b^2}$, $h_{a2b} = \sqrt{a^2 + 4b^2}$, $h_{2ab} = \sqrt{4a^2 + b^2}$ and $\lambda_m = m!!/(m+1)!!$,

$$I_m(\text{Vc}) = \frac{\lambda_m a^{m+3} b}{(m+2)(m+3)} \left[(2^{m+4} + 4) \operatorname{asinh} \frac{b}{a} - 2^{m+4} \operatorname{asinh} \frac{b}{2a} - 4 \operatorname{asinh} \frac{2b}{a} \right] + (a \leftrightarrow b) \\ + \frac{2}{(m+2)(m+3)(m+4)} \left[-(2^{m+3} - 1)(a^{m+4} + b^{m+4}) \right. \\ \left. - (2^{m+3} + 2) \hat{G}_m(a, b) + \hat{G}_m(a, 2b) + \hat{G}_m(2a, b) \right], \quad (22)$$

where $\hat{G}_m$ is a universal “hypotenuse polynomial” in $h = \sqrt{a^2 + b^2}$ and $s = a^2 b^2$,

$$\hat{G}_{-1} = h^3, \quad \hat{G}_1 = (h^4 - 5s)h, \quad \hat{G}_3 = (h^6 - \frac{49}{8}sh^2)h, \quad \hat{G}_5 = (h^8 - \frac{123}{16}sh^4 + \frac{45}{8}s^2)h, \quad (23)$$

the $m = 7$ member having been confirmed out of sample. Specializing,

$$I_{-1}(\text{Vc}) = 2a^2 b \left[3 \operatorname{asinh} \frac{b}{a} - 2 \operatorname{asinh} \frac{b}{2a} - \operatorname{asinh} \frac{2b}{a} \right] + 2ab^2 \left[3 \operatorname{asinh} \frac{a}{b} - 2 \operatorname{asinh} \frac{a}{2b} - \operatorname{asinh} \frac{2a}{b} \right] \\ - (a^3 + b^3) - 2h_{ab}^3 + \frac{1}{3}(h_{a2b}^3 + h_{2ab}^3), \quad (24)$$

$$I_1(\text{Vc}) = a^4 b \left[\frac{3}{2} \operatorname{asinh} \frac{b}{a} - \frac{4}{3} \operatorname{asinh} \frac{b}{2a} - \frac{1}{6} \operatorname{asinh} \frac{2b}{a} \right] + (a \leftrightarrow b) \\ - \frac{1}{2}(a^5 + b^5) - \frac{3}{5}h_{ab}^5 + 3s h_{ab} + \frac{1}{30}(h_{a2b}^5 + h_{2ab}^5) - \frac{2}{3}s(h_{a2b} + h_{2ab}), \quad (25)$$

$$I_3(\text{Vc}) = a^6 b \left[\frac{33}{20} \operatorname{asinh} \frac{b}{a} - \frac{8}{5} \operatorname{asinh} \frac{b}{2a} - \frac{1}{20} \operatorname{asinh} \frac{2b}{a} \right] + (a \leftrightarrow b) \\ - \frac{3}{5}(a^7 + b^7) - \frac{22}{35}h_{ab}^7 + \frac{77}{20}s h_{ab}^3 + \frac{1}{105}(h_{a2b}^7 + h_{2ab}^7) - \frac{7}{30}s(h_{a2b}^3 + h_{2ab}^3), \quad (26)$$

with $s = a^2 b^2$. Equation (24) is (22) at $m = -1$ term by term; the two were derived independently and the agreement of the $\lambda_{-1} = 1$, $(m+2)(m+3) = 2$ specialization with the discovered coefficients is one of the reasons to trust the general formula. The atom set of Vc has six members and never changes with m ; only the coefficients do, and they are homogeneous of degree $m+3$.

Verification of the coefficients. All 259 rational coefficients of the explicit Vc, Vp and Xg forms at $m = -1, 1, 3$ were recovered independently from pairMom data by solving a linear system in the conjectured atom basis, with residuals between 3.9×10^{-30} and 1.8×10^{-53} : Vc needs 14, 17 and 20 terms at those three m . This is a genuinely independent confirmation of the algebra, since the coefficients come from numbers, not from the derivation.

Cancellation. As written, (22) loses up to 11.7 digits at $b/a = 10^{\pm 6}$, and the top-level amplification explains all of it: there are four algebraic buckets $a^{m+4} + b^{m+4}$, h_{ab} , h_{a2b} , h_{2ab} , each of size $O(a^{m+4})$, against an integral of size $O(a^{m+2}b^2)$; the asinh block is already a second difference and is fine. The shipped code does not use (22): Vc's four boxes are integrated directly, and its four box types include the doubly shifted fall₁/fall₁ that motivated the whole of Section 5.

Measured. Correctness 2.43×10^{-45} (sliver, $m = -1$), 144-pair sweep 1.37×10^{-48} , cross-check against the polar form 9.01×10^{-90} and against the discovery form 6.95×10^{-90} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
Vc	0.08	0.00	0.04	0.00	0.17	0.17	0.00	0.17	0.26	0.05	0.13	0.23	0.19	0.23

worst 0.26 at $(10^{-1}, 10^{-6})$, $m = 7$; the best-conditioned code in the whole set, which is worth contrasting with the 11.7 digits of the explicit form. Against Gila: **wekV** entries 1–3 agree with the series to 4.98×10^{-10} at **intOrd** 48 and 8.99×10^{-11} at 64. Gila has no closed form for Vc, so there is no $m = -1$ static-part check; the series is the first exact value this geometry has had.

3.5 P1 and P2: parallel, aligned



Left, P1 at $\Delta = 0$: yzU of the target against yzL of the source, one cell length apart. Right, P2 at $\Delta = (1, 0, 0)$, two cell lengths apart. Panels $A = [0, a] \times [0, b] \times \{0\}$, $B = [0, a] \times [0, b] \times \{c\}$ or $\{2c\}$.

These are S with a frozen offset. The assembly is the first row of (15) with $c = s_3$ and $c = 2s_3$, and the closed form is (17) together with (8): the c -powers and the atan term of Φ are the whole difference from the self face. Explicitly, for every integer $m \geq -1$,

$$I_m(\text{P1}) = 4 \mathcal{R}[0, a; 0, b; (a, -1), (b, -1)] \text{ at } c = s_3, \quad I_m(\text{P2}) = \text{the same at } c = 2s_3. \quad (27)$$

Both are separated pairs: r never vanishes on the domain, so there is no singular behaviour to arrange around, and the difficulty is purely one of conditioning.

Cancellation. These two geometries are where the naive route fails worst. Written through (8), the P family loses up to 29 digits, and the culprit is M_{11} alone: its numerator $R^p - d_A^p - d_B^p + c^p$ with $p = m + 4$ is a four-term difference in which, when $c \gg a, b$, all four terms are within $O((a^2 + b^2)/c^2)$ of c^p while the true value is $p(p-2)a^2b^2c^{p-4}/4$, so about $2 \log_{10}(c/a) + 2 \log_{10}(c/b)$ digits vanish.

Measured, M_{11} loses 28.4 digits at $(a, b, c, m) = (0.01, 1, 10^6, -1)$, and the assembled P1 loses *exactly* the same 28.4 at the same point: the whole P-family loss is M_{11} and nothing else. M_{00} by contrast loses 0.9 digits over the entire grid. The shipped code removes this completely by never forming M_{11} : it integrates each box against its own weight with the ladder of Section 5.2, and for boxes far from the origin — which is exactly the regime $c \gg a, b$ — it uses the centred Taylor expansion.

Measured. Correctness 4.50×10^{-60} (P1, fatZ, $m = 1$) and 3.84×10^{-60} (P2, fatX, $m = 11$); note that these are three orders of magnitude below the touching geometries, because **pairMom** converges much faster on a separated pair. 144-pair sweep 1.12×10^{-75} (P1) and 9.31×10^{-76} (P2). Cross-checks against the polar explicit forms 1.26×10^{-84} and 2.24×10^{-83} , the worst at fatZ, $m = -1$, where the explicit form is itself cancelling.

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
P1	0.48	0.00	0.29	0.12	0.41	0.24	0.36	0.45	0.42	0.44	0.46	0.53	0.60	0.53
P2	0.62	0.00	0.20	0.23	0.19	0.24	0.29	0.39	0.37	0.55	0.47	0.55	0.60	0.59

worst 0.60 (P1) at $(10^{-2}, 10^{-3})$, $m = 11$ and 0.62 (P2) at $(10^{-6}, 10^{-1})$, $m = -1$ — against 28.4 and 25.4 for the plain closed form. Against Gila: the order-9 fixed rule reproduces the series to 8.9×10^{-14} (P1) and 1.1×10^{-15} (P2) at a $(1/32)^3$ cell, and P1 is the one code where that rule fails outright on a slender cell, at 1.57×10^{-2} (Section 7).

3.6 P1s and P2s: parallel, laterally shifted



Left, P1s at $\Delta = (1, 0, 0)$: xzL of the target against xzU of the source. Right, P2s at $\Delta = (1, 1, 0)$: yzU against yzL. Panels $A = [0, a] \times [0, b] \times \{0\}$, $B = [a, 2a] \times [0, b] \times \{c\}$ or $\{2c\}$.

Ef with a frozen offset: one in-plane axis carries the shifted triangle of the flat edge, the other the centred triangle of the self face. The assembly is the middle row of (15) at $c = s_3$ and $c = 2s_3$, and the closed form is again (17)–(8).

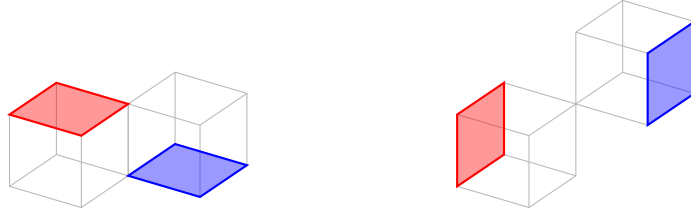
Cancellation. Two mechanisms combine here, and P1s was the geometry that exposed the second. As for P1/P2, M_{11} costs up to 28.3 digits as written. On top of that, the shifted box $[a, 2a]$ carries the corner amplification $u_2/(u_2 - u_1) = 2$ per shifted axis, which was measured exactly: for the fall/fall box type the ratio $|u_2 v_2 M_{00}|/|W|$ tends to $u_2 v_2/(h_u h_v)$, equal to 4 for a corner box, 8 with one shifted axis and 16 with two. With the closed forms alone P1s and P2s stalled at 1.15 and 1.27 digits at $m = -1$, above the bar; the centred Taylor branch of Section 5.2 brings them to 0.64 and 0.63.

Measured. Correctness 4.53×10^{-60} (P1s, fatZ, $m = 9$) and 4.01×10^{-60} (P2s, small, $m = 7$); 144-pair sweep 9.44×10^{-76} and 4.72×10^{-76} ; cross-checks against the polar forms 5.27×10^{-85} and 5.60×10^{-84} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
P1s	0.64	0.00	0.24	0.17	0.32	0.27	0.42	0.43	0.42	0.54	0.53	0.54	0.54	0.62
P2s	0.63	0.00	0.17	0.25	0.32	0.25	0.41	0.41	0.54	0.55	0.54	0.54	0.61	0.63

worst 0.64 (P1s) at $(10^{-2}, 10^{-1})$, $m = -1$ and 0.63 (P2s) at $(10^4, 10^{-4})$, $m = 12$. Against Gila’s order-9 rule at a $(1/32)^3$ cell: 2.1×10^{-14} (P1s) and 5.7×10^{-15} (P2s); on the slender cell P1s degrades to 2.81×10^{-4} while P2s stays at 4.3×10^{-15} , which is the expected pattern (the failure is driven by the *gap* being the short edge, and P2s’s gap is two short edges).

3.7 P1s-b and P2s-b: parallel, doubly shifted



Left, P1s-b at $\Delta = (1, 1, 0)$: xyL of the target against xyU of the source. Right, P2s-b at $\Delta = (1, 1, 1)$: yzU against yzL. Panels $A = [a, 2a] \times [b, 2b] \times \{0\}$ against $B = [0, a] \times [0, b] \times \{c\}$, and the analogue at $2c$.

Both in-plane axes are shifted, where P1s and P2s shift one and leave the other fully overlapping. Structurally these are *Vc with a frozen offset*: the assembly is the third row of (15) at $c = s_3$ and $c = 2s_3$, and the four boxes are the four quadrants of two shifted triangles. Table 1 of the companion document does not distinguish them from P1s and P2s, which is why they carry the “-b” suffix.

These two sub-codes are the only ones in the set with *no independent explicit closed form*. The polar corner moments G_{ij} already carry c , so extending (15)’s Vc row to $c \neq 0$ is immediate in principle, but no separate explicit routine was written or cross-checked, and table (c) of the verification accordingly reports “no explicit form” for them. Their evidence is instead: the 320-bit reference cache at eleven shapes (table (a)), the 144-pair sweep against **pairMom** (table (b)), homogeneity, orientation invariance under all six axis permutations, and the $a \leftrightarrow b$ symmetry — which they possess, at 1.33×10^{-96} and 8.65×10^{-97} , and P1s and P2s do *not* (0.99 and 0.84). That last check is the discriminating one: it shows the shifted codes are not being silently confused with their unshifted parents.

Cancellation. Their $\text{fall}_1/\text{fall}_1$ box is the worst-conditioned box type in the two-dimensional set, amplification exactly 16, and with closed forms only they lost 1.24 and 1.38 digits at $m = -1$. The centred Taylor branch covers that box type in *every* cell of the grid, at most 42 levels deep with condition number 1.26, and brings them to 0.34 and 0.28.

Measured. Correctness 4.85×10^{-60} (P1s-b, fatZ, $m = 1$) and 4.46×10^{-60} (P2s-b, fatX, $m = 1$); 144-pair sweep 6.42×10^{-76} and 5.97×10^{-76} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
P1s-b	0.34	0.00	0.17	0.14	0.14	0.29	0.31	0.44	0.45	0.34	0.43	0.51	0.60	0.55
P2s-b	0.28	0.00	0.13	0.13	0.14	0.18	0.37	0.38	0.38	0.54	0.50	0.58	0.58	0.60

worst 0.60 (P1s-b) at $(10^0, 10^{-3})$, $m = 11$ and 0.60 (P2s-b) at $(10^{-3}, 10^{-3})$, $m = 12$. Against Gila's order-9 rule at a $(1/32)^3$ cell, 9.9×10^{-15} and 1.9×10^{-15} ; on the slender cell, 1.80×10^{-7} and 1.28×10^{-8} .

4 The perpendicular geometries

Six sub-codes are three-dimensional in the sense of (1): the two panels lie in perpendicular planes, so only the shared axis admits a difference variable and carries a trapezoid, while the two normal axes keep their own coordinates and contribute indicators. They are Ec, Vp, Xg, Xg-b, Xg-c and Xg-d. These are the hard cases, and they were attacked by three families independently.

4.1 The master reduction

Put the shared axis along s , with half-width a and triangle centre $\delta \in \{0, \pm a\}$, and let the two normal axes carry the indicator ranges $[\beta_1, \beta_2]$ and $[\gamma_1, \gamma_2]$, each of which is $[0, L]$ (the panels' projections meet) or $[L, 2L]$ (they are a cell apart). With $\Psi_m(u; B, C) = \int_0^B \int_0^C (u^2 + v^2 + w^2)^{m/2}$ and P_m of (12),

$$I_m = \sum_{i,j \in \{1,2\}} (-1)^{i+j} W_m(\beta_i, \gamma_j), \quad W_m(B, C) = \begin{cases} 2P_m(a; B, C), & \delta = 0, \\ P_m(2a; B, C) - 2P_m(a; B, C), & \delta = \pm a, \end{cases} \quad (28)$$

with $W_m(0, \cdot) = W_m(\cdot, 0) = 0$. This single formula covers every perpendicular face pair of Table 1 including the Xg variants with a shifted weight and both ranges offset, which need all eight P_m terms. The named orientations are

$$\begin{aligned} \text{Ec}/x &: 2P_m(a; b, c), & \text{Ec}/y &: 2P_m(b; a, c), \\ \text{Vp}/x &: P_m(2b; a, c) - 2P_m(b; a, c), & \text{Vp}/y &: P_m(2a; b, c) - 2P_m(a; b, c), \\ \text{Xg}/y &: 2[P_m(a; 2b, c) - P_m(a; b, c)], & \text{Xg}/x &: 2[P_m(b; 2a, c) - P_m(b; a, c)], \end{aligned} \quad (29)$$

and Xg-b, Xg-c, Xg-d are (28) with $(\delta = \pm a, \text{ one range offset})$, $(\delta = 0, \text{ both offset})$ and $(\delta = \pm a, \text{ both offset})$ respectively.

The primitives are the ladder of Section 2.2. Recorded here in the form the box-potential route wrote them, with $\rho = \sqrt{t^2 + B^2}$, $\sigma = \sqrt{t^2 + C^2}$, $R = \sqrt{t^2 + B^2 + C^2}$:

$$\begin{aligned} \Psi_n(t; B, C) &= \frac{B J_n(C; \rho) + C J_n(B; \sigma) + n t^2 \Psi_{n-2}}{n+2}, & \Psi_0 &= BC, \\ \Psi_{-1}(t; B, C) &= B \operatorname{asinh} \frac{C}{\rho} + C \operatorname{asinh} \frac{B}{\sigma} - t \operatorname{atan} \frac{BC}{tR}, \\ K^{(m)}(A, B, C) &= \frac{A \Psi_m(A; B, C) + B \Psi_m(B; A, C) + C \Psi_m(C; A, B)}{m+3}, \\ P_m(T; B, C) &= T K^{(m)}(T, B, C) - \frac{\Psi_{m+2}(T; B, C) - \Psi_{m+2}(0; B, C)}{m+2}. \end{aligned} \quad (30)$$

Unrolled at odd order, with $G_j = B J_{2j-1}(C; \rho) + C J_{2j-1}(B; \sigma)$,

$$\Psi_{2q-1}(t; B, C) = \frac{t^{2q} \Psi_{-1} + \sum_{j=1}^q t^{2(q-j)} G_j}{2q+1}, \quad (31)$$

the first three G_j being, with $\alpha = \text{asinh}(C/\rho)$ and $\beta = \text{asinh}(B/\sigma)$,

$$\begin{aligned} G_1 &= \frac{1}{2}(B\rho^2\alpha + C\sigma^2\beta) + BCR, \\ G_2 &= \frac{3}{8}(B\rho^4\alpha + C\sigma^4\beta) + \frac{3}{8}BCR(\rho^2 + \sigma^2) + \frac{1}{2}BCR^3, \\ G_3 &= \frac{5}{16}(B\rho^6\alpha + C\sigma^6\beta) + \frac{5}{16}BCR(\rho^4 + \sigma^4) + \frac{5}{24}BCR^3(\rho^2 + \sigma^2) + \frac{1}{3}BCR^5. \end{aligned} \quad (32)$$

The corner box moment at the first two odd orders, with $\alpha_c = \text{asinh}(c/h_{ab})$ and its mirrors and $\tau_a = \text{atan}(bc/(aR))$ and its mirrors, is

$$K^{(1)} = \frac{1}{12} \left[2ab h_{ab}^2 \alpha_c + 2ac h_{ac}^2 \alpha_b + 2bc h_{bc}^2 \alpha_a + 3abcR - a^4 \tau_a - b^4 \tau_b - c^4 \tau_c \right], \quad (33)$$

$$\begin{aligned} K^{(3)} &= \frac{1}{30} \left[ab \left(\frac{5}{4} h_{ab}^4 + a^4 + b^4 \right) \alpha_c + ac \left(\frac{5}{4} h_{ac}^4 + a^4 + c^4 \right) \alpha_b + bc \left(\frac{5}{4} h_{bc}^4 + b^4 + c^4 \right) \alpha_a \right. \\ &\quad \left. + 4abcR^3 - a^6 \tau_a - b^6 \tau_b - c^6 \tau_c \right]. \end{aligned} \quad (34)$$

Verification of the master formula on its own terms, before any assembly: all 24 perpendicular face pairs at each of four offsets, cell $(3/2, 1, 2/5)$, $m = -1, \dots, 8$ — 960 values, maximum relative error 1.53×10^{-51} against `pairMom`.

4.2 Ec: perpendicular, sharing an edge



Left, Ec within one cell: xzL and xyL of the same cell. Right, Ec across cells at $\Delta = (1, 1, 0)$: yzL of the target against xzU of the source. Panels $A = [0, a] \times [0, b] \times \{0\}$, $B = [0, a] \times \{b\} \times [0, c]$; shared edge a , widths b and c .

The shared axis carries the centred triangle $(a - |u|)$, so $\delta = 0$ and both indicator ranges start at zero: (28) collapses to a single term, $I_m = 2P_m(a; b, c)$, and the boxed formula is (12) with (30). Collected at the first three odd orders, with $\Delta_b = \text{asinh}(c/b) - \text{asinh}(c/h_{ab})$ and $\Delta_c = \text{asinh}(b/c) - \text{asinh}(b/h_{ac})$,

$$P_{-1}(a; b, c) = \frac{b^3}{6} \Delta_b + \frac{c^3}{6} \Delta_c + \frac{a^2 b}{2} \alpha_c + \frac{a^2 c}{2} \alpha_b + abc \alpha_a - \frac{a^3}{6} \tau_a - \frac{ab^2}{2} \tau_b - \frac{ac^2}{2} \tau_c - \frac{a^2 bc}{3(h_{bc} + R)}, \quad (35)$$

$$\begin{aligned} P_1(a; b, c) &= \frac{1}{120} \left[3b^5 \Delta_b + 3c^5 \Delta_c + 5a^2 b(a^2 + 2b^2) \alpha_c + 5a^2 c(a^2 + 2c^2) \alpha_b + 20abc h_{bc}^2 \alpha_a \right. \\ &\quad \left. - 2a^5 \tau_a - 10ab^4 \tau_b - 10ac^4 \tau_c + 19a^2 bcR - 7bc(R^3 - h_{bc}^3) \right], \end{aligned} \quad (36)$$

$$\begin{aligned} P_3(a; b, c) &= \frac{1}{1680} \left[15b^7 \Delta_b + 15c^7 \Delta_c + 7a^2 b(3a^4 + 5a^2 b^2 + 9b^4) \alpha_c + 7a^2 c(3a^4 + 5a^2 c^2 + 9c^4) \alpha_b \right. \\ &\quad \left. + 14abc(5h_{bc}^4 + 4b^4 + 4c^4) \alpha_a - 8a^7 \tau_a - 56ab^6 \tau_b - 56ac^6 \tau_c \right. \\ &\quad \left. + bc(172a^2 R^3 - 66a^4 R - 15[R(h_{ab}^4 + h_{ac}^4) - h_{bc}(b^4 + c^4)]) - 26(R^5 - h_{bc}^5) \right]. \end{aligned} \quad (37)$$

Equation (35) is `rSrfEdgCrn` (equation (23) of the companion document) term for term, which is how the identity (12) was first confirmed: the grouping of the b^3 and c^3 logarithms into Δ_b and Δ_c , which the companion document found by hand and called the essential step, falls out of (12) automatically. At $m = -1$ the new form and `rSrfEdgCrn` agree to 1.2×10^{-88} .

Ec is symmetric in $b \leftrightarrow c$, and that symmetry is what **wekE**'s two-orientation average relies on. It was computed, not assumed: 3.8×10^{-96} over all shapes and all m , and correspondingly the two orientations Gila averages give bit-identical closed-form moments (relative difference exactly 0 at all four cell scales).

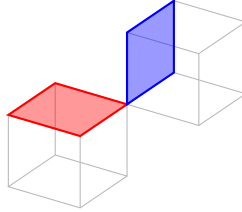
Cancellation. The entire loss of the naive route is the difference $\Psi_{m+2}(T) - \Psi_{m+2}(0)$ in (12), whose amplification reaches 10^{11} to 10^{23} . Propagating that difference by its own recurrence, rather than forming the two Ψ 's and subtracting, takes Ec from 15.7 digits lost to 1.2–1.3; the shipped code goes further and never performs the $TK - \Delta\Psi$ split at all, integrating the weighted box directly by $\nabla \cdot (fXr^m)$ with $f = q - u$ (Section 5.5), which removes the fixed factor-3 amplification of that split.

Measured. Correctness 7.35×10^{-44} (fatX, $m = -1$) — the worst entry of the whole correctness table, and it is the reference's floor, not the formula's: recomputing **pairMom** at three increasing order sets at this shape gives 1.46×10^{-22} , 2.64×10^{-30} , 7.35×10^{-44} . 144-pair sweep 4.84×10^{-49} ; cross-check against the master formula 1.16×10^{-95} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
Ec	0.17	0.01	0.15	0.41	0.30	0.28	0.43	0.46	0.49	0.54	0.72	0.67	0.74	0.77

worst 0.77 at $(10^{-3}, 10^0)$, $m = 12$. Ec is also the code that first exceeds the one-digit bar above $m = 12$: 1.15 digits at $m = 25$ and 1.28 at $m = 30$ (Section 5.8). Against Gila: **wekE** entries 7–9 agree with the series to 9.29×10^{-11} at **intOrd** 48 and 6.06×10^{-12} at 64; I_{-1} against $4\pi f^2$ **rSrfEdgCrn** is 2.54×10^{-16} .

4.3 Vp: perpendicular, sharing only a corner



$\Delta = (1, 1, 1)$: yzL of the target against xyU of the source. Panels $A = [0, a] \times [0, b] \times \{0\}$, $B = \{a\} \times [b, 2b] \times [0, c]$: the two panels touch only at the point $(a, b, 0)$.

The shared axis now carries a shifted triangle, so $\delta = \pm a$ and $W_m = P_m(2a; \cdot) - 2P_m(a; \cdot)$; both indicator ranges still start at zero, so the double sum still collapses to one term. Nothing was known about this geometry before. Alongside the master formula there is an explicit form for each m , obtained by the discovery route and independently confirmed coefficient by coefficient. With $h_{ab}, h_{a2b}, h_{ac}, h_{bc}, h_{2bc}$ as before, $R = \sqrt{a^2 + b^2 + c^2}$ and $R_2 = \sqrt{a^2 + 4b^2 + c^2}$,

$$\begin{aligned}
I_{-1}(\text{Vp}) = & -\frac{a^3}{6} \operatorname{asinh} \frac{c}{a} - \frac{c^3}{6} \operatorname{asinh} \frac{a}{c} + \left(\frac{a^3}{3} - ab^2\right) \operatorname{asinh} \frac{c}{h_{ab}} + \left(2ab^2 - \frac{a^3}{6}\right) \operatorname{asinh} \frac{c}{h_{a2b}} \\
& + \left(\frac{c^3}{3} - b^2c\right) \operatorname{asinh} \frac{a}{h_{bc}} + \left(2b^2c - \frac{c^3}{6}\right) \operatorname{asinh} \frac{a}{h_{2bc}} + 2abc \left[\operatorname{asinh} \frac{2b}{h_{ac}} - \operatorname{asinh} \frac{b}{h_{ac}} \right] \\
& + a^2b \left[\operatorname{atan} \frac{bc}{aR} - \operatorname{atan} \frac{2bc}{aR_2} \right] + bc^2 \left[\operatorname{atan} \frac{ab}{cR} - \operatorname{atan} \frac{2ab}{cR_2} \right] \\
& + \frac{b^3}{3} \operatorname{atan} \frac{ac}{bR} - \frac{4b^3}{3} \operatorname{atan} \frac{ac}{2bR_2} + \frac{ac}{3} (2R - R_2 - h_{ac}), \tag{38}
\end{aligned}$$

$$I_1(\text{Vp}) = -\frac{a^5}{40} \operatorname{asinh} \frac{c}{a} - \frac{c^5}{40} \operatorname{asinh} \frac{a}{c} + \frac{a(3a^4 - 10a^2b^2 - 5b^4)}{60} \operatorname{asinh} \frac{c}{h_{ab}} - \frac{a(3a^4 - 40a^2b^2 - 80b^4)}{120} \operatorname{asinh} \frac{c}{h_{a2b}}$$

$$\begin{aligned}
& -\frac{c(5b^4+10b^2c^2-3c^4)}{60} \operatorname{asinh} \frac{a}{h_{bc}} + \frac{c(80b^4+40b^2c^2-3c^4)}{120} \operatorname{asinh} \frac{a}{h_{2bc}} \\
& + \frac{abc(a^2+c^2)}{3} \left[\operatorname{asinh} \frac{2b}{h_{ac}} - \operatorname{asinh} \frac{b}{h_{ac}} \right] + \frac{a^4b}{6} \left[\operatorname{atan} \frac{bc}{aR} - \operatorname{atan} \frac{2bc}{aR_2} \right] + \frac{bc^4}{6} \left[\operatorname{atan} \frac{ab}{cR} - \operatorname{atan} \frac{2ab}{cR_2} \right] \\
& + \frac{b^5}{30} \operatorname{atan} \frac{ac}{bR} - \frac{8b^5}{15} \operatorname{atan} \frac{ac}{2bR_2} + \frac{ac(7a^2-12b^2+7c^2)}{60} R - \frac{ac(7a^2-48b^2+7c^2)}{120} R_2 - \frac{7ac(a^2+c^2)}{120} h_{ac}, \quad (39) \\
I_3(\text{Vp}) = & -\frac{a^7}{112} \operatorname{asinh} \frac{c}{a} - \frac{c^7}{112} \operatorname{asinh} \frac{a}{c} + \frac{a(15a^6-63a^4b^2-35a^2b^4-21b^6)}{840} \operatorname{asinh} \frac{c}{h_{ab}} \\
& - \frac{a(15a^6-252a^4b^2-560a^2b^4-1344b^6)}{1680} \operatorname{asinh} \frac{c}{h_{a2b}} - \frac{c(21b^6+35b^4c^2+63b^2c^4-15c^6)}{840} \operatorname{asinh} \frac{a}{h_{bc}} \\
& + \frac{c(1344b^6+560b^4c^2+252b^2c^4-15c^6)}{1680} \operatorname{asinh} \frac{a}{h_{2bc}} + \frac{abc(9a^4+10a^2c^2+9c^4)}{60} \left[\operatorname{asinh} \frac{2b}{h_{ac}} - \operatorname{asinh} \frac{b}{h_{ac}} \right] \\
& + \frac{a^6b}{15} \left[\operatorname{atan} \frac{bc}{aR} - \operatorname{atan} \frac{2bc}{aR_2} \right] + \frac{bc^6}{15} \left[\operatorname{atan} \frac{ab}{cR} - \operatorname{atan} \frac{2ab}{cR_2} \right] + \frac{b^7}{105} \operatorname{atan} \frac{ac}{bR} - \frac{64b^7}{105} \operatorname{atan} \frac{ac}{2bR_2} \\
& + \frac{ac}{840} (41a^4 - 90a^2b^2 + 52a^2c^2 - 50b^4 - 90b^2c^2 + 41c^4) R \\
& - \frac{ac}{1680} (41a^4 - 360a^2b^2 + 52a^2c^2 - 800b^4 - 360b^2c^2 + 41c^4) R_2 - \frac{ac(41a^4+52a^2c^2+41c^4)}{1680} h_{ac}. \quad (40)
\end{aligned}$$

The atom-coefficient laws. The atom set never changes with m — six atoms for Vc, fourteen for Vp, sixteen for Xg — and only the coefficients move, homogeneously of degree $m+3$. For Vp, with $\lambda_m = m!!/(m+1)!!$,

$$\boxed{
\begin{aligned}
\operatorname{asinh} \frac{c}{a} &\mapsto -\frac{\lambda_{m+2} a^{m+4}}{(m+2)(m+4)}, \\
\operatorname{atan} \frac{bc}{aR} &\mapsto \frac{2b a^{m+3}}{(m+2)(m+3)}, & \operatorname{atan} \frac{2bc}{aR_2} &\mapsto -\frac{2b a^{m+3}}{(m+2)(m+3)}, \\
\operatorname{atan} \frac{ac}{bR} &\mapsto \frac{2b^{m+4}}{(m+2)(m+3)(m+4)}, & \operatorname{atan} \frac{ac}{2bR_2} &\mapsto -\frac{2^{m+4} b^{m+4}}{(m+2)(m+3)(m+4)},
\end{aligned}
} \quad (41)$$

together with the $a \leftrightarrow c$ mirrors. These reproduce the explicit forms above at $m = -1, 1, 3$ and were confirmed out of sample.

Symmetries. Vp is symmetric in $a \leftrightarrow c$ (8.97×10^{-97}) and is *not* symmetric in $b \leftrightarrow c$ ($3.67 \times 10^{+2}$) or $a \leftrightarrow b$ ($7.32 \times 10^{+2}$). The absence matters as much as the presence: Gila's **wekV** averages two orientations of each Vp entry, and the two are the same integral only because of the $a \leftrightarrow c$ symmetry — which the closed forms confirm by giving bit-identical values for the two orientations at all four cell scales.

Cancellation. The explicit form loses up to 22.6 digits in **Float64**, the worst of any geometry, at $b/a = 10^{-6}$, $c/a = 10^6$; the three responsible groups are $\frac{ac}{3}(2R - R_2 - h_{ac})$ and the two second differences $\operatorname{asinh} \frac{a}{c} - 2 \operatorname{asinh} \frac{a}{h_{bc}} + \operatorname{asinh} \frac{a}{h_{2bc}}$ and its $a \leftrightarrow c$ mirror. The shipped code uses none of these expressions.

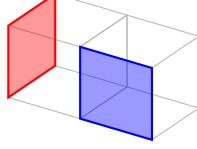
Measured. Correctness 3.41×10^{-44} (fatX, $m = -1$; reference floor), 144-pair sweep 2.57×10^{-49} , cross-check against the master formula 1.03×10^{-95} and against the discovery form 9.00×10^{-91} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
Vp	0.46	0.00	0.21	0.06	0.23	0.18	0.35	0.23	0.49	0.42	0.50	0.47	0.61	0.73

worst 0.73 at $(10^5, 10^5)$, $m = 12$. Against Gila: **wekV** entries 4–6 agree with the series to 1.80×10^{-9} at **intOrd** 48 and 3.26×10^{-10} at 64 — the largest discrepancy anywhere in the comparison, and it

is Gila's error, since it appears on the slender cell where DIRECTFN's transformations produce sliver triangles.

4.4 Xg: perpendicular planes that do not meet



$\Delta = (1, 0, 0)$: xzL of the target against yzL of the source. Panels $A = [0, a] \times [0, b] \times \{0\}$, $B = [0, a] \times \{2b\} \times [0, c]$: perpendicular planes, one cell apart.

The shared axis carries the centred triangle ($\delta = 0$) and exactly one indicator range is offset to $[b, 2b]$, so (28) gives the two-term $I_m = 2[P_m(a; 2b, c) - P_m(a; b, c)]$. Explicitly, with the same hypotenuse names,

$$\begin{aligned} I_{-1}(\text{Xg}) = & \frac{8b^3}{3} \operatorname{asinh} \frac{c}{2b} - \frac{b^3}{3} \operatorname{asinh} \frac{c}{b} + \frac{c^3}{3} \left[\operatorname{asinh} \frac{2b}{c} - \operatorname{asinh} \frac{b}{c} \right] + 2abc \left[2 \operatorname{asinh} \frac{a}{h_{2bc}} - \operatorname{asinh} \frac{a}{h_{bc}} \right] \\ & + (a^2c - \frac{c^3}{3}) \left[\operatorname{asinh} \frac{2b}{h_{ac}} - \operatorname{asinh} \frac{b}{h_{ac}} \right] + (2a^2b - \frac{8b^3}{3}) \operatorname{asinh} \frac{c}{h_{a2b}} - (a^2b - \frac{b^3}{3}) \operatorname{asinh} \frac{c}{h_{ab}} \\ & + \frac{a^3}{3} \left[\operatorname{atan} \frac{bc}{aR} - \operatorname{atan} \frac{2bc}{aR_2} \right] + ab^2 \operatorname{atan} \frac{ac}{bR} - 4ab^2 \operatorname{atan} \frac{ac}{2bR_2} + ac^2 \left[\operatorname{atan} \frac{ab}{cR} - \operatorname{atan} \frac{2ab}{cR_2} \right] \\ & + \frac{2bc}{3} (R - h_{bc}) - \frac{4bc}{3} (R_2 - h_{2bc}), \end{aligned} \quad (42)$$

$$\begin{aligned} I_1(\text{Xg}) = & \frac{8b^5}{5} \operatorname{asinh} \frac{c}{2b} - \frac{b^5}{20} \operatorname{asinh} \frac{c}{b} + \frac{c^5}{20} \left[\operatorname{asinh} \frac{2b}{c} - \operatorname{asinh} \frac{b}{c} \right] \\ & + abc \left[\frac{2(4b^2+c^2)}{3} \operatorname{asinh} \frac{a}{h_{2bc}} - \frac{b^2+c^2}{3} \operatorname{asinh} \frac{a}{h_{bc}} \right] + \frac{c(5a^4+10a^2c^2-3c^4)}{60} \left[\operatorname{asinh} \frac{2b}{h_{ac}} - \operatorname{asinh} \frac{b}{h_{ac}} \right] \\ & + \frac{b(5a^4+40a^2b^2-48b^4)}{30} \operatorname{asinh} \frac{c}{h_{a2b}} - \frac{b(5a^4+10a^2b^2-3b^4)}{60} \operatorname{asinh} \frac{c}{h_{ab}} + \frac{a^5}{30} \left[\operatorname{atan} \frac{bc}{aR} - \operatorname{atan} \frac{2bc}{aR_2} \right] \\ & + \frac{ab^4}{6} \operatorname{atan} \frac{ac}{bR} - \frac{8ab^4}{3} \operatorname{atan} \frac{ac}{2bR_2} + \frac{ac^4}{6} \left[\operatorname{atan} \frac{ab}{cR} - \operatorname{atan} \frac{2ab}{cR_2} \right] \\ & - \frac{bc(12a^2-7b^2-7c^2)}{60} R + \frac{bc(12a^2-28b^2-7c^2)}{30} R_2 + \frac{7bc(4b^2+c^2)}{30} h_{2bc} - \frac{7bc(b^2+c^2)}{60} h_{bc}, \end{aligned} \quad (43)$$

$$\begin{aligned} I_3(\text{Xg}) = & \frac{16b^7}{7} \operatorname{asinh} \frac{c}{2b} - \frac{b^7}{56} \operatorname{asinh} \frac{c}{b} + \frac{c^7}{56} \left[\operatorname{asinh} \frac{2b}{c} - \operatorname{asinh} \frac{b}{c} \right] \\ & + abc \left[\frac{144b^4+40b^2c^2+9c^4}{30} \operatorname{asinh} \frac{a}{h_{2bc}} - \frac{9b^4+10b^2c^2+9c^4}{60} \operatorname{asinh} \frac{a}{h_{bc}} \right] \\ & + \frac{c(21a^6+35a^4c^2+63a^2c^4-15c^6)}{840} \left[\operatorname{asinh} \frac{2b}{h_{ac}} - \operatorname{asinh} \frac{b}{h_{ac}} \right] \\ & + \frac{b(21a^6+140a^4b^2+1008a^2b^4-960b^6)}{420} \operatorname{asinh} \frac{c}{h_{a2b}} - \frac{b(21a^6+35a^4b^2+63a^2b^4-15b^6)}{840} \operatorname{asinh} \frac{c}{h_{ab}} \\ & + \frac{a^7}{105} \left[\operatorname{atan} \frac{bc}{aR} - \operatorname{atan} \frac{2bc}{aR_2} \right] + \frac{ab^6}{15} \operatorname{atan} \frac{ac}{bR} - \frac{64ab^6}{15} \operatorname{atan} \frac{ac}{2bR_2} + \frac{ac^6}{15} \left[\operatorname{atan} \frac{ab}{cR} - \operatorname{atan} \frac{2ab}{cR_2} \right] \\ & - \frac{bc}{840} (50a^4 + 90a^2b^2 + 90a^2c^2 - 41b^4 - 52b^2c^2 - 41c^4) R \\ & + \frac{bc}{420} (50a^4 + 360a^2b^2 + 90a^2c^2 - 656b^4 - 208b^2c^2 - 41c^4) R_2 \\ & + \frac{bc(656b^4+208b^2c^2+41c^4)}{420} h_{2bc} - \frac{bc(41b^4+52b^2c^2+41c^4)}{840} h_{bc}, \end{aligned} \quad (44)$$

with the general- m atom coefficients

$$\begin{array}{ll}
\operatorname{asinh} \frac{c}{2b} \mapsto \frac{2^{m+5} \lambda_{m+2} b^{m+4}}{(m+2)(m+4)}, & \operatorname{asinh} \frac{c}{b} \mapsto -\frac{2 \lambda_{m+2} b^{m+4}}{(m+2)(m+4)}, \\
\operatorname{atan} \frac{bc}{aR} \mapsto \frac{2a^{m+4}}{(m+2)(m+3)(m+4)}, & \operatorname{atan} \frac{ac}{bR} \mapsto \frac{2a b^{m+3}}{(m+2)(m+3)}, \\
\operatorname{atan} \frac{ac}{2bR_2} \mapsto -\frac{2a(2b)^{m+3}}{(m+2)(m+3)}, & \operatorname{atan} \frac{ab}{cR} \mapsto \frac{2a c^{m+3}}{(m+2)(m+3)}.
\end{array} \tag{45}$$

Xg has no symmetry at all: $a \leftrightarrow b$, $a \leftrightarrow c$ and $b \leftrightarrow c$ all give order-unity or larger relative differences (4.74×10^2 , 1.49×10^{-1} , 8.60×10^2), and the only exception is the trivial $m = 0$ case where $I_0 = a^2bc$.

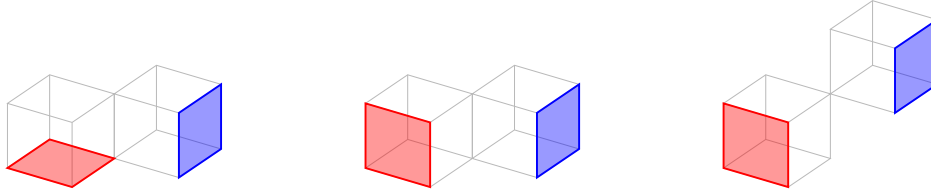
Cancellation. The explicit form loses up to 12.8 digits at $b/a = 10^6$ through three groups: $\frac{2bc}{3}(R - h_{bc}) - \frac{4bc}{3}(R_2 - h_{2bc})$, and the two pairs of asinh differences that mirror each other in $b \leftrightarrow 2b$. The box route avoids all three.

Measured. Correctness 3.84×10^{-60} (fatZ, $m = 7$) — three orders better than the touching geometries, again because **pairMom** converges faster on a non-touching pair; 144-pair sweep 1.24×10^{-74} ; cross-check against the master formula 2.50×10^{-95} and against the discovery form 9.23×10^{-90} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
Xg	0.45	0.00	0.35	0.36	0.39	0.46	0.47	0.54	0.59	0.59	0.64	0.65	0.61	0.73

worst 0.73 at $(10^4, 10^5)$, $m = 12$. Against Gila’s order-9 rule, 1.4×10^{-14} at a $(1/32)^3$ cell and 8.3×10^{-15} on the slender cell.

4.5 Xg-b, Xg-c, Xg-d: the three further perpendicular classes



Left, Xg-b at $\Delta = (1, 1, 0)$: yzU against xyL, the shared-axis triangle shifted and the indicators at (gap 1, gap 0). Middle, Xg-c at $\Delta = (1, 1, 0)$: yzU against xzL, triangle centred but both indicators a full cell away. Right, Xg-d at $\Delta = (1, 1, 1)$: yzU against xzL, triangle shifted and both indicators a cell away.

Table 1 calls all three Xg. They are three different integrals. In the master formula (28):

- **Xg-b** has $\delta = \pm a$ and one offset range, so $I_m = 2[P_m(2a; \cdot) - 2P_m(a; \cdot)]$ evaluated at the two ends of the offset range and differenced: four P_m terms. Its reduced integral is *exactly* Vp’s under the right argument map, a coincidence found independently by the discovery route (“Xg variant (ii) has exactly Vp’s reduced integral”) and by the signature classification, which puts 20 of the 144 pairs in this class — the third largest, after Ec and Vp.
- **Xg-c** has $\delta = 0$ and both ranges offset: four P_m terms, all with the unshifted weight. It is symmetric in $b \leftrightarrow c$ (9.05×10^{-97}), which canonical Xg is not; this is the “both indicators a full cell away” symmetry, and it was found by the symmetry sweep rather than anticipated.

- **Xg-d** has $\delta = \pm a$ and both ranges offset: the full eight P_m terms of (28), the most shifted member of the family. It too is symmetric in $b \leftrightarrow c$ (8.09×10^{-97}).

The canonical panels used throughout are

$$\text{Xg-b} : A = \{2b\} \times [a, 2a] \times [0, c], \quad B = [0, b] \times [0, a] \times \{0\}, \quad (46)$$

$$\text{Xg-c} : A = \{2b\} \times [c, 2c] \times [0, a], \quad B = [0, b] \times \{0\} \times [0, a], \quad (47)$$

$$\text{Xg-d} : A = \{2b\} \times [c, 2c] \times [a, 2a], \quad B = [0, b] \times \{0\} \times [0, a]. \quad (48)$$

None of the three has a collected explicit form of its own; the master formula is the closed form, and the box route is the evaluation.

Measured. Correctness 4.40×10^{-60} (Xg-b, sliver, $m = 9$), 4.06×10^{-60} (Xg-c, cube, $m = 3$), 4.75×10^{-60} (Xg-d, fatX, $m = 1$); 144-pair sweep 2.09×10^{-74} , 5.77×10^{-75} , 5.26×10^{-75} ; cross-checks against the master formula 2.22×10^{-95} , 3.85×10^{-95} , 8.73×10^{-95} .

m	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
Xg-b	0.78	0.00	0.38	0.35	0.44	0.44	0.57	0.54	0.60	0.59	0.67	0.63	0.75	0.77
Xg-c	0.90	0.00	0.37	0.38	0.50	0.54	0.59	0.74	0.66	0.71	0.74	0.76	0.91	0.78
Xg-d	0.85	0.00	0.14	0.31	0.18	0.21	0.37	0.38	0.43	0.43	0.59	0.56	0.68	0.66

Xg-c carries the global worst of the whole digits table, 0.91 at $(a/b, c/b) = (10^5, 10^0)$, $m = 11$; Xg-b's worst is 0.78 at $(10^{-2}, 10^1)$, $m = -1$ and Xg-d's is 0.85 at $(10^{-1}, 10^0)$, $m = -1$. Xg-c is also the second code to cross the bar above $m = 12$ (1.11 digits at $m = 25$). Against Gila's order-9 rule at a $(1/32)^3$ cell: 4.1×10^{-15} , 2.7×10^{-15} and 7.3×10^{-15} ; on the slender cell Xg-b degrades to 1.37×10^{-6} while Xg-c and Xg-d stay at 5.7×10^{-15} and 7.2×10^{-15} .

5 The cancellation-free arrangement

Deriving the closed forms took one round. Arranging them so that `Float64` keeps fifteen of its sixteen digits took three, and produced the only genuinely new mathematics in this work. This section is that story, told in the order in which the obstructions appeared.

5.1 The rules, and the diagnostic

The rules are the companion document's, and they held up:

1. **Never subtract a length from a hypotenuse.** $\sqrt{x^2 + y^2} - x = y^2/(\sqrt{x^2 + y^2} + x)$, and $h^n - b^n = \frac{a^2}{h+b} \sum_{i=0}^{n-1} h^i b^{n-1-i}$, applied recursively to any polynomial in h, p, q that vanishes as $p \rightarrow 0$.
2. **Write differences of asinh as $\log(1+x)$** with x a ratio of positive quantities obtained by rationalizing.
3. **Keep atan arguments as ratios of positive quantities**, never $\text{atan}(1/x)$ or acot of something that changes sign.
4. **Find the exact identities first**, using the amplification $\sum_i |\text{term}_i| / |\sum_i \text{term}_i|$ in extended precision as the diagnostic: if it explains the digits lost, the loss is between top-level terms and the fix is to group them; if the top-level amplification is one and digits are still lost, the loss is inside a sub-expression.
5. **Make symmetric integrals symmetric in form.**
6. **Extended precision only as a last resort.**

To these the work added a seventh, which turned out to matter more than any of them: *measure before regrouping, and measure the amplification of the arrangement you are proposing before you derive it*. Twice, an arrangement that was obviously better on paper was measurably worse (Section 5.3), and once a claimed impossibility dissolved when the right quantity was measured.

The metric throughout is $\log_{10}(|x_{64} - x_{\text{big}}|/(\epsilon |x_{\text{big}}|))$ with $\epsilon = 2.22 \times 10^{-16}$ and x_{big} the 320-bit evaluation of the same expression at the *same* `Float64` inputs, converted exactly (`BigFloat(a64)`, never `BigFloat(10)^i`). Getting that last detail wrong shows an input mismatch of $(m+4)/2$ roundings as if it were a loss of the formula.

5.2 The two-dimensional box

`pairBxs` produces exactly three per-axis shapes: `ris0` (weight u on $[0, p]$), `fal0` (weight $p - u$ on $[0, p]$) and `fal1` (weight $2p - u$ on $[p, 2p]$), so nine box types in two dimensions. The one-dimensional ladder carries all three weights at once,

$$\begin{aligned} (m+1) J_1^{(m)} &= t_2 \phi_m(t_2) - t_1 \phi_m(t_1) + m d^2 J_1^{(m-2)}, \\ (m+2) J_{\text{ris}}^{(m)} &= (t_2 - t_1) t_2 \phi_m(t_2) - t_1 J_1^{(m)} + m d^2 J_{\text{ris}}^{(m-2)}, \\ (m+2) J_{\text{fal}}^{(m)} &= t_2 J_1^{(m)} - (t_2 - t_1) t_1 \phi_m(t_1) + m d^2 J_{\text{fal}}^{(m-2)}, \end{aligned} \quad (49)$$

running upward from $m = -1$ (odd) or $m = 0$ (even), with the squared offset threaded through the whole computation so that an offset is rounded once and never re-squared.

The $m = -1$ seeds are where the work is. With $A_i = \sqrt{t_i^2 + d^2}$,

$$\zeta = \operatorname{asinh} \frac{t_2}{d} - \operatorname{asinh} \frac{t_1}{d} = \log \frac{t_2 + A_2}{t_1 + A_1} = \log(1 + x), \quad x = \frac{(t_2 - t_1) + (t_2^2 - t_1^2)/(A_2 + A_1)}{t_1 + A_1}, \quad (50)$$

taken as `log1p(x)` when $x < 1/2$ and as the logarithm of the ratio itself otherwise, and $H = A_2 - A_1 = (t_2 - t_1)(t_2 + t_1)/(A_2 + A_1)$. Then $J_1^{(-1)} = \zeta$, $J_{\text{ris}}^{(-1)} = H - t_1 \zeta$, and the new piece,

$$\boxed{J_{\text{fal}}^{(-1)} = h \zeta + C, \quad C = 2(t_1 \cosh \delta + A_1 \sinh \delta)(\delta \cosh \delta - \sinh \delta), \quad \delta = \frac{\zeta}{2},} \quad (51)$$

with $h = (t_2 - t_1)/2$ and $\delta \cosh \delta - \sinh \delta = \sum_{k \geq 1} 2k \delta^{2k+1}/(2k+1)!$, every term positive. This is the centred first moment $\int_{t_1}^{t_2} (t_m - t)/\sqrt{t^2 + d^2}$ written as a manifestly positive product; the naive $t_2 \zeta - H$ has amplification $2t_2/(t_2 - t_1)$, which is 4 for a $[p, 2p]$ box. The representation (51) costs 2δ in relative error through the $e^{2\delta}$ growth of its factors, so it is used only for $\zeta \leq 2$; for a $[p, 2p]$ box $\delta \leq \frac{1}{2} \log 2 = 0.347$ always, so it governs exactly where it is needed.

Offset differences get their own recurrence rather than being formed by subtraction. With $d_h^2 - d_\ell^2 = \Delta d^2$ passed exactly,

$$\begin{aligned} 2D_1 &= t_2(A_{2h} - A_{2\ell}) - t_1(A_{1h} - A_{1\ell}) + [d_h^2 J_1^h - d_\ell^2 J_1^\ell], \\ 3D_{\text{ris}} &= (t_2 - t_1)t_2(A_{2h} - A_{2\ell}) - t_1 D_1 + [\dots], \quad 3D_{\text{fal}} = t_2 D_1 - (t_2 - t_1)t_1(A_{1h} - A_{1\ell}) + [\dots], \end{aligned} \quad (52)$$

with $A_{ih} - A_{i\ell} = \Delta d^2/(A_{ih} + A_{i\ell})$, and the bracket written either as $d_h^2 J^h - d_\ell^2 J^\ell$ or as $d_h^2 D^{(-1)} + \Delta d^2 J^\ell$ — the same expression algebraically, but they cancel in complementary regimes, so the one with the smaller term sum is taken at each call. The $m = -1$ offset difference is again a $\log(1 + x)$ of a manifestly positive rationalization.

The $m = -1$ two-dimensional seed on a corner box is the rectangle potential Φ , regrouped so that its leading behaviour is one positive product,

$$\Phi = pq\left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{R}\right) + p\left[\operatorname{asinh} \frac{q}{\alpha} - \frac{q}{\alpha}\right] + q\left[\operatorname{asinh} \frac{p}{\beta} - \frac{p}{\beta}\right] + c[x - \operatorname{atan} x], \quad x = \frac{pq}{cR}, \quad (53)$$

with $\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{R} = \frac{1}{\beta} + \frac{q^2}{\alpha R(R+\alpha)}$ positive and the two bracketed differences summed as their own alternating series for arguments below $1/2$. On a shifted box the four-corner atan difference $\Delta_u \Delta_v \operatorname{atan}(uv/(cR))$ is taken as the best conditioned of three exact forms — the plain four-corner difference, the u -first nested form, and the v -first nested form — because the two nested forms degenerate in complementary regimes and never together. Choosing correctly took that object from 11.78 digits lost to 0.62.

5.3 Why the elementary closed forms cannot reach the bar

With all of the above, one box type remained above one digit: `fall/fall`, the doubly shifted box, at 1.83 digits, and with it the assembled P1s, P2s, P1s-b and P2s-b at 1.15 to 1.38. The mechanism is forced. Every exact elementary form of

$$W(\text{fall}, \text{fall}) = \iint (u_2 - u)(v_2 - v) \frac{du dv}{R} = u_2 v_2 M_{00} - u_2 M_{01} - v_2 M_{10} + M_{11} \quad (54)$$

has those four coefficients and no others: an antiderivative of $(u_2 - u)(v_2 - v)/R$ is fixed up to $f(u) + g(v)$, which drops out of the double difference, and of the four moments only M_{00} carries the atan , so they are functionally independent and cannot be recombined. Hence for a slowly varying kernel $W \rightarrow h_u h_v M_{00}$ and

$$\frac{|u_2 v_2 M_{00}|}{|W|} \rightarrow \frac{u_2 v_2}{h_u h_v} = \begin{cases} 4 & \text{corner box,} \\ 8 & \text{one shifted axis,} \\ 16 & \text{two shifted axes,} \end{cases} \quad (55)$$

and these were measured to attain their theoretical values exactly (4.0, 8.0, 8.0, 16.0 on the grid). The first round concluded from this that 1.16 digits was a floor.

That conclusion was wrong, and the way it was shown wrong is worth recording. The centred decomposition

$$W = h_u h_v M_{00} - h_v X - h_u Y + Z, \quad X = \iint \frac{p}{R}, \quad Y = \iint \frac{q}{R}, \quad Z = \iint \frac{pq}{R}, \quad (56)$$

with $p = u - u_m$, $q = v - v_m$, has $O(1)$ coefficients, so its amplification with *exact* X, Y, Z is 1.0 — measured, exactly 1.0 over the whole grid. Forming X, Y, Z as differences of the M_{ij} gives 9.0, and $9 + 3 + 3 + 1 = 16$ recovers the direct form's factor, which is why the naive centring gains nothing. So the question is whether X, Y, Z have accurate elementary forms of their own. They have elementary forms: with $\alpha_i^2 = u_i^2 + c^2$, $\zeta(v) = \operatorname{asinh}(u_2/\beta) - \operatorname{asinh}(u_1/\beta)$, D_v the offset difference of $\operatorname{asinh}(v/\alpha_i)$ and T the four-corner atan difference,

$$X = \frac{1}{2} D_v [v(A_2 - A_1)] + \frac{h_u^2 + c^2 - u_m^2}{2} D_v - u_m D_v [v \zeta(v)] + u_m c T, \quad (57)$$

$$Z = \Delta \Delta \left[R \left(\frac{u^2 + v^2 + c^2}{3} - \frac{u_m u + v_m v}{2} \right) \right] + v_m \frac{u_1 u_2 - c^2}{2} D_v + u_m \frac{v_1 v_2 - c^2}{2} D_u - u_m v_m c T, \quad (58)$$

both verified against the definitions to 4.2×10^{-72} and 2.7×10^{-48} . The collapse that produces X is not obvious: the $\operatorname{asinh}(v/\alpha_i)$ terms of M_{10} and of $u_m M_{00}$ carry the coefficient $\alpha_i^2/2 - u_m u_i =$

$((u_i - u_m)^2 + c^2 - u_m^2)/2$, and $(u_i - u_m)^2 = h_u^2$ for both i , so the two pieces merge into a single offset difference with a constant coefficient.

And they are useless. In the far regime each of the four terms of X is $O(u_m h_u h_v / c)$ while X itself is $O(u_m h_u^3 h_v / c^3)$, so the closed form cancels by $(c/h_u)^2$: the measured $\max |\text{term}|/|X|$ is 1.2×10^{25} on the grid, *identical* to the naive route's $u_m M_{00}/|X|$, and for Z both are 4.8×10^{37} . The cancellation is a property of the elementary antiderivative, not of the arrangement, and forming X, Y, Z from their own closed forms is worse than naive by six orders of magnitude (1.94×10^6 against 9.0). Two other routes to the same objects — carrying the one-dimensional centred primitive (51) into the second integration, and re-associating the mixed second difference of the antiderivative — end at exactly these expressions, because they are the only elementary ones there are. So: *no elementary closed form of the doubly shifted seed reaches one digit*, and the 1.16 figure is a real statement about closed forms. It is not a statement about the problem.

5.4 The Taylor series about the box centre

What breaks the deadlock is that the doubly shifted box is, in exactly the regime where its closed form is ill-conditioned, *small compared with its distance to the singularity*. There R^m is analytic on a polydisc containing the box, and the moment has an exact, finitely truncated expansion in which every coefficient is elementary:

$$R^m = (\rho^2 + 2u_m p + 2v_m q + p^2 + q^2)^{m/2} = \sum_{i,j} c_{ij} p^i q^j, \quad \rho^2 = u_m^2 + v_m^2 + c^2, \quad (59)$$

$$W_{ab} = \sum_{i,j} c_{ij} \mu_i^{(a)} \nu_j^{(b)}, \quad \mu_i^{(a)} = \int_{-h}^h w_a(p) p^i dp, \quad (60)$$

with the three weight moments exact and one line each,

$$\mu_i^{(1)} = \frac{2h^{i+1}}{i+1} \text{ (i even), } 0 \text{ (i odd);} \quad \mu_i^{(2,3)} = \frac{2h^{i+2}}{i+1} \text{ (i even), } \pm \frac{2h^{i+2}}{i+2} \text{ (i odd),} \quad (61)$$

the sign being + for the rise weight $p + h$ and − for the fall weight $h - p$; all nine (a, b) combinations share one coefficient table. The coefficients come from $g \partial_p f = \frac{m}{2} f \partial_p g$ with $f = g^{m/2}$ and its mirror in q :

$$\begin{aligned} \rho^2(i+1)c_{i+1,j} &= (m-2i)u_m c_{ij} + (m-i+1)c_{i-1,j} - 2v_m(i+1)c_{i+1,j-1} - (i+1)c_{i+1,j-2}, \\ \rho^2(j+1)c_{0,j+1} &= (m-2j)v_m c_{0j} + (m-j+1)c_{0,j-1}, \quad c_{00} = \rho^m. \end{aligned} \quad (62)$$

Every entry of level $n = i + j$ needs only levels $n - 1$ and $n - 2$, so the table is built one anti-diagonal at a time and stopped when the series has converged.

This is not quadrature. No sample of the integrand is ever taken; the expression is an exact identity truncated at a level whose remainder the series itself controls, in the same sense that the ladder (4) is exact and the double-factorial sum (5) is finite. It is an evaluation scheme for a closed form, not a replacement for one, and it is used only where the closed forms are ill-conditioned. The honest way to state the deliverable is therefore: the closed forms of Sections 3 and 4 are the answer; the shipped evaluator computes them by whichever of two exact routes is better conditioned at the point asked for.

Five implementation details decide whether it works.

- **Prescale the lengths by a power of two** (`ldexp`, `exact`) so that $\rho = O(1)$. Without it $\mu_i = 2h^{i+1}/(i+1)$ overflows around level 180 whenever $h > 1$ while c_{ij} underflows, and the two garbage factors multiply. The unscaling is per entry, because W_{ab} has homogeneity degree $m+2+\deg w_a+\deg w_b$, not $m+4$; getting that wrong cost 27 digits at $(a/b, c/b) = (10^{-6}, 10^6)$, measured.
- **Add the level sums smallest first.** In the natural order the branch lost 1.2 digits at 75 levels by random-walk accumulation; reversed, 0.2 at the same depth.
- **Require two consecutive small levels** for convergence, because for the $(1, 1)$ weight pair every odd level vanishes identically.
- **Gate the attempt:** the polydisc parameter $\lambda > 1/2$, where λ is the positive root of $(h_u^2 + h_v^2)\lambda^2 + 2(u_m h_u + v_m h_v)\lambda - \rho^2 = 0$ (this rejects corner boxes at $c \approx 0$, where the origin sits on the box corner and no expansion converges); a projection from the geometric-mean level ratio at $n \geq 10$ that abandons an attempt needing more than 140 levels; and a final guard $\sum |\text{terms}| \leq 4|\sum|$. A rejection falls back to the closed forms, and the region where it happens is exactly the region where the closed forms amplify by 4 rather than 16.
- **Re-seed every rung of the ladder** from the series, not just $m = -1$: its domain of convergence does not depend on m , so the ladder's own six-step accumulation disappears.

This took the `fal1/fal1` box at $m = 11$ from 1.04 to 0.74.

Because the tolerance is $\epsilon(T)$ -based, at `BigFloat` precision the series needs about seven times more levels and is mostly rejected — correctly, since the closed forms are exact at that precision. It is one code path, generic in T .

Measured coverage of the doubly shifted box, the one whose closed form amplifies by 16: *every* cell of the 13×14 grid, at most 42 levels deep, condition number 1.259. Coverage falls to 70/182 only for the corner box, whose closed form amplifies by 4 and needs no help. The series branch checked on its own against 320-bit closed forms, over every used weight pair, every converged cell and every $m = -1, \dots, 12$: worst 0.61 digits, that is 4ϵ . The effect on the box types and the geometries:

box type	closed only	hybrid	geometry	closed only	hybrid
<code>fal0/fal0</code>	0.90	0.60	S	0.40	0.40
<code>fal0/fal1</code>	1.40	0.91	Ef	0.49	0.37
<code>fal0/ris0</code>	0.63	0.63	Vc	0.73	0.26
<code>fal1/fal0</code>	1.46	0.95	P1	0.90	0.60
<code>fal1/fal1</code>	1.83	0.77	P2	0.88	0.87
<code>fal1/ris0</code>	0.90	0.83	P1s	1.15	0.62
<code>ris0/fal0</code>	0.55	0.55	P2s	1.27	0.63
<code>ris0/fal1</code>	0.88	0.70	P1s-b	1.24	0.60
<code>ris0/ris0</code>	0.60	0.60	P2s-b	1.37	0.60
global	1.83	0.95	global	1.37	0.87

At $m = -1$ alone the `fal1/fal1` box goes from 1.83 to 0.12. The cost is a factor 1.3 to 6.9 over the closed forms on the boxes where the series is used, which is accepted deliberately: these are build-time integrals evaluated once per cell shape.

5.5 The three-dimensional box

The perpendicular geometries needed a different set of identities, because their boxes are three-dimensional and the weight sits on one axis only. Three ideas carry it.

Shifted-box divergence identities. Every primitive is written directly on the shifted range $[lo, hi]$ rather than as a difference of corner-box values, so the geometry-level inclusion–exclusion disappears entirely: every box enters the assembly with a positive weight. In one dimension,

$$(n+1)J_n[lo, hi; d] = hi S_h^n - lo S_\ell^n + n d^2 J_{n-2}, \quad hi S_h^n - lo S_\ell^n = \Delta S_h^n + lo \Delta (hi + lo) \frac{\Sigma(S_h, S_\ell, n)}{S_h + S_\ell}, \quad (63)$$

with $\Delta = hi - lo$ and $\Sigma(A, B, n) = \sum_{i=0}^{n-1} A^i B^{n-1-i}$, and $J_{-1} = \log(1+x)$ with $x = [\Delta + (hi^2 - lo^2)/(S_h + S_\ell)]/(lo + S_\ell)$ positive. The negative orders that the thin branch needs have their own positive seed, $J_{-3} = (hi^2 - lo^2)/(S_h S_\ell (hi S_\ell + lo S_h))$.

The δ -grouping. Every opposite-face pair in a divergence identity has the form $hi f(hi) - lo f(lo)$, which is re-associated as $\Delta f(hi) + lo [f(hi) - f(lo)]$ with the bracket computed by its own cancellation-free offset-difference recurrence rather than by subtraction. This single change took the worst assembled loss from 1.31 to 1.03 digits, and it is worth 0.3 to 0.4 digits everywhere downstream.

The solid angle. The $m = -1$ seed of the shifted rectangle moment is $\Psi_{-1} = \text{bdy}(-1) - t\Omega$ with Ω the solid angle subtended by $[a_1, a_2] \times [b_1, b_2]$ at height t . Computing Ω as a four-corner difference of arctangents cancels; computing it by the van Oosterom–Strackee formula on the two triangles,

$$\Omega = \sum_{2 \text{ triangles}} 2 \operatorname{atan} \frac{\text{num}}{\text{den}}, \quad \text{num} = t(a_2 - a_1)(b_2 - b_1), \quad \text{den} = |P_1||P_2||P_3| + \sum (P_i \cdot P_j)|P_k|, \quad (64)$$

keeps every quantity positive in the positive octant, so there is no arctangent differencing anywhere in the three-dimensional path.

The weighted box without a split. Rather than $P_m = TK^{(m)} - \Delta\Psi/(m+2)$, which carries a fixed amplification of about 3, the weighted box is integrated by its own divergence identity: with $f = q - u$, $\nabla \cdot (fXr^m) = [(m+4)f - q]r^m$, and with $f = u - p$, $\nabla \cdot (fXr^m) = [(m+4)f + p]r^m$, so the near faces contribute their own terms instead of being differenced away.

Two binomial branches. Where a box is far from the origin along one axis ($4(hi_j^2 + hi_k^2) \leq lo_d^2$) the face terms of the divergence identity are $O(\text{dist})$ larger than the integral, so the expansion $I = \sum_n \binom{m/2}{n} \sum_i \binom{n}{i} X_{2i} Y_{2(n-i)} Z_{m-2n}$ in the polynomial moments is used instead and the face terms never appear. Where a box is thin on one axis ($hi_i^2 \leq (lo_j^2 + lo_k^2)/10^4$) the mirror expansion in the small variable is used, whose term n carries weight $\leq 10^{-4n}$ and therefore needs only $16 - 4n$ digits, which is what the negative-order recurrences deliver.

The $1/(p+1) - 1/(p+2)$ trap. The polynomial moment $\int_{lo}^{hi} u^p (c + du) du$ evaluated by its naive antiderivative computes $hi^{p+2}[1/(p+1) - 1/(p+2)]$ as written and loses $\log_{10}(4p)$ digits; this cost 1.19 digits at $m = 9$ until it was found. It is expanded about the near end instead,

$$\int_{lo}^{hi} u^p (c + du) du = \sum_j \binom{p}{j} lo^{p-j} \left[w_{lo} \frac{\Delta^{j+1}}{j+1} + d \frac{\Delta^{j+2}}{j+2} \right] \quad (d \geq 0), \quad (65)$$

with $w_{lo} = c + dlo \geq 0$, term by term positive, and the mirror form for $d < 0$. The same trap sits one level up in the two-dimensional weighted moment and is routed around the same way.

What each stage was worth. Measured, worst assembled I_m over the 13×13 grid at odd m :

stage	worst assembled I_m	fall L0L box	fall LLL box
shifted-box divergence forms, faces not grouped	1.31	1.43	1.61
+ δ -grouped Ψ face pairs	1.03	1.17	1.35
+ separated and thin branches, before the <code>polMom</code> fix	1.19	1.09	1.19
final	0.89	1.02	0.93

For comparison, the corner-box-plus-inclusion-exclusion route on the same grid, in its own cancellation-free variant, ranges from 0.15 to 1.86 digits and exceeds one digit for every geometry at high m ; the shifted-box route is what made the bar reachable.

The residual, named exactly. One box type sat at 1.02 digits in the three-dimensional development: `fall|L0L`, the compact box $[1, 2] \times [0, 1] \times [1, 2]$ with weight $(2 - u)$ at $m = -1$, at $a = b = c$, that is at the *most benign* geometry, the unit cube. The diagnosis is complete and structural rather than a regrouping that was missed. Two amplifications survive every rearrangement: $\Psi_{-1} = \text{bdy} - t\Omega$ has $\sum |\text{terms}|/|\text{total}| \rightarrow 3$ exactly for a rectangle at a height comparable to its own size (measured 2.70), because the boundary part is $\sim 2A/t$ and $t\Omega \sim A/t$; and the weighted rectangle moment $B_{-1} = q\Psi_{-1} - \iint a/r$ has amplification $\geq a_2/(\Delta a/2) = 4$ when $a_1 = a_2/2$ (measured 6.71), because the weight has mean $\Delta a/2$ over the box while every unweighted moment carries the scale a_2 . Their product with the K level, $2.70 \times 1.63 \times 1.65 = 7.3$, is a floor of 0.86 digits for that box, and 1.02 is that floor plus the ordinary rounding of the thirty elementary operations that build the terms. Three things would remove it, none of them a regrouping: a cancellation-free $\Delta\Psi_n(t_0, t_1)$ primitive, whose $n \geq 1$ steps are routine but whose seed needs $t_1\Omega(t_1) - t_0\Omega(t_0)$ without cancellation and that identity was not found; a genuinely second-order weighted primitive $\int (q - u)^2 r^m$ via $\nabla \cdot ((q - u)^2 Xr^m) = (m + 5)(q - u)^2 r^m - 2q(q - u)r^m$, at the price of a whole new family with its own seeds; or double-double evaluation of the $m = -1$ seeds alone. In the shipped file, assembled through the canonical panels, the worst three-dimensional entry is 0.91 (`Xg-c` at $m = 11$), so this box type never dominates a geometry.

5.6 Sliver boxes and exact breakpoints

A box whose two breakpoints on some axis differ by one unit in the last place is a disaster for every divergence identity: each face term is $hi/\Delta \sim 10^{16}$ times the integral. Such boxes are not produced by an exact decomposition — `pairBox` on the lattice only ever makes $hi = 2lo$ or $lo = 0$ — but they are produced abundantly when the breakpoint arithmetic `sort(unique([ $p - s, q - s, p - r, q - r$ ]))` is done in `Float64` and two mathematically equal breakpoints round apart. The three-dimensional integrator’s positivity sweep found them: 49 of 8528 grid boxes, producing wrong-sign garbage worth up to 2% of the assembled I_m .

The shipped fix is upstream and exact. `pairBxs` converts the panel coordinates to `Rational{BigInt}` first (exact for any binary float), forms the breakpoints, the piece weights and the frozen offsets exactly, and rounds to T once at the end. Two breakpoints that coincide mathematically are then *identical*, never an ulp apart, and the spurious sliver cannot be produced by the decomposition at all.

What the exact path cannot fix is a sliver already present in the input panels. Gila’s own face coordinates are computed as $\text{off} \cdot s + \text{sig} \cdot s/2$, and $3s/2$ is not exactly representable, so a `Float64` panel is one ulp away from the exact one and the exact decomposition faithfully reproduces that. Two remedies exist and both are supported: hand `facePair` a `Rational` s , or pass `tol`, which merges breakpoints closer than that relative tolerance. Measured at $s = (0.1, 0.3, 0.7)$, which is

inexact on every axis, over all 144 face pairs and $m = -1, \dots, 12$: 544 boxes on the exact route (96 of them caught by the thin-axis guard) against 384 with `tol` = 8ϵ , and worst relative errors 2.11×10^{-15} and 1.92×10^{-15} respectively — both correct, the tolerance route simply avoiding the slivers rather than surviving them.

The same ulp effect is why one measurement protocol matters: building the `Float64` box list from `Float64 panels` rather than from the exact panels moves six entries of the three-dimensional digits table by up to 0.11 digits, because the two evaluations then integrate slightly different geometries. `canonPanels` only ever multiplies by two and is exact, which is why the published grids are unaffected.

5.7 The off-lattice thin-indicator branch

The adversarial audit found the one real conditioning failure of the three-dimensional integrator, and it is off the lattice. For a perpendicular pair whose *indicator* axis is narrow relative to its offset, the divergence identities amplify exactly as the code’s own comment predicts, and the original guard fired four orders of magnitude too late: with one narrow axis the loss reached 3.96 digits just above the gate, and with both indicator axes narrow — the guard returned only the first — it reached 6.90 and kept growing, confirmed independently against `pairMom` at 6.9 digits lost. The two-dimensional integrator has no such problem: the same experiment stays under 0.28 digits down to a ratio of 1.5×10^{-8} .

The fix has three parts. The gate becomes $2(hi - lo) < hi$, *strict*, so a lattice box with $hi = 2lo$ sits exactly on the boundary and never enters the branch — which is what keeps every published table bitwise unchanged. Both thin axes are handled, not one. And the branch itself uses a precision-adaptive Gauss–Legendre rule along the thin axis against the two-dimensional closed form at the node heights, with the order taken from the analyticity radius,

$$n = \text{clamp}\left(\left\lceil \frac{-\log \epsilon(T)}{2 \log \varrho} \right\rceil + 3, 3, 240\right), \quad \varrho = \frac{m_d + \sqrt{m_d^2 - h_f^2}}{h_f} \geq 3 + \sqrt{8}, \quad (66)$$

the bound following from the gate; that is order 11 to 14 in `Float64` and about 66 at 320 bits, and the nodes are generated by Newton iteration on the Legendre recurrence to $\epsilon(T)$, generic in T . Measured, the worst loss over the sweep falls from 3.96 to 0.84 digits with one narrow axis and from 7.61 to 0.90 with two, with zero cells above the bar; against `pairMom` directly, from 6.9 digits lost to 0.06. The earlier three-point rule also capped `BigFloat` accuracy at about 10^{-24} once the gate fired, a 59-order discontinuity; the adaptive rule removes it, landing within 5 to 20 units of the working precision at 128, 256 and 320 bits alike, flat across the gate.

Two honest statements about this branch. First, *it is* a quadrature, unlike everything else in the file, and it is therefore not part of the closed-form claim; it is an evaluation device for geometries the lattice never produces. Second, *it is never reached for the fifteen lattice sub-geometries*: every lattice box has $hi = 2lo$ or $lo = 0$, so the strict gate excludes it, and the regression check confirms this the hard way — all 39 438 values of the digits grid and all eleven reference shapes are bitwise identical before and after the branch was added.

5.8 Above $m = 12$

The one-digit property is a statement about $m \leq 12$. Measured on a coarse grid ($a/b, c/b \in \{10^{-6}, 10^{-3}, 1, 10^3, 10^6\}$, plus $c = 0$ for the two-dimensional codes):

code	$m=13$	15	17	19	21	23	25	worst cell
Ec	0.85	0.91	0.97	1.01	1.08	1.10	1.15	$(10^{-6}, 10^0)$
Xg-c	0.76	0.79	0.91	0.98	1.04	1.05	1.11	$(10^0, 10^0)$
Vp	0.60	0.76	0.72	0.79	0.88	0.89	0.96	$(10^6, 10^6)$
Xg	0.55	0.75	0.80	0.85	0.89	0.91	0.96	$(10^6, 10^6)$
P2s	0.76	0.86	0.81	0.84	0.94	0.96	0.99	$(10^{-6}, 10^{-6})$
S	0.10	0.30	0.01	0.16	0.31	0.27	0.07	$(10^{-3}, 10^{-6})$
Vc	0.00	0.00	0.26	0.16	0.09	0.22	0.09	$(10^0, 10^{-6})$

(global worst 1.15, 11 cells above one digit of 1785; at $m = 30$ the global worst is 1.28, again Ec.) The mechanism is neither cancellation nor assembly: it is the one-dimensional ladder **jSeg**. Every term of a rung is positive, so nothing cancels; the loss is one rounding per rung, accumulated. Measured on $[1, 2]$ over 200 offsets:

m	-1	1	5	9	13	17	21	25	31
rungs	0	1	3	5	7	9	11	13	16
digits lost	0.07	0.03	0.56	0.74	0.94	1.04	1.13	1.20	1.28
$\log_{10}(\text{rungs})$	—	0.00	0.48	0.70	0.85	0.95	1.04	1.11	1.20

so the law is $\text{loss} = \log_{10}(\text{rungs}) + 0.08$ with $\text{rungs} = (m + 1)/2$, and Ψ , K and the assembled moment inherit it. At $m \leq 12$ that is six rungs and a bound of 0.78, which is why the published table stays at 0.91.

The two-dimensional side escapes this because the centred series re-seeds every rung, so it owns an m -independent evaluation. The three-dimensional side has none for the general box shape (the separated branch covers only the dominant-axis regime), so removing the accumulation means writing a new integrator, not making a local change; a compensated ladder would also do it but would move every **Float64** value in the published tables. The behaviour is documented in the file rather than fixed, and it is listed in Section 9 as an open item. Correctness is entirely unaffected: at $m = 30$ the worst error against **pairMom** over all fifteen sub-codes is 8.9×10^{-61} , while the reference's own order-to-order gap at the same entry is 7.5×10^{-46} .

5.9 The pitfalls, collected

Each of these was found by measurement, not by inspection, and each cost real digits. They are collected here because the next reader of this code will meet them again.

1. **The log1p branch.** $\log(1 + x)$ is only the right form while $x < 1/2$; when the ratio is far from one, $1 - x$ is the small quantity and the logarithm of the ratio must be taken directly. Getting this backwards cost 11.99 digits per box at $a/b = 10^6$; branching correctly gives 2.03 at the same point.
2. **Thread d^2 , never d .** Computing $\sqrt{t^2 + d^2}$ from a d that was itself a square root costs $m/2$ roundings at exponent m : about 0.4 digits at $m = 11$.
3. **The grouping of an offset difference.** The bracket $d_h^2 D + \Delta d^2 J^\ell$ and $d_h^2 J^h - d_\ell^2 J^\ell$ are algebraically the same and cancel in complementary regimes; choose per call by term sum. The three-dimensional analogue is sharper: $d_0^2 D J_{n-2} + e^2 J_{n-2}(d_1)$ keeps the $n = 1$ step finite as $d_0 \rightarrow 0$ while $d_1^2 D J + e^2 J(d_0)$ subtracts two $\log(1/d_0)$ -sized quantities. In the polar route the same trap reads: write $J_{K-2}(q; \alpha) + c^2 \delta_{K-2}$, never $\alpha^2 \delta_{K-2} + J_{K-2}(q; c)$, which at $K = 1$, $c \ll p$ subtracts two divergent $\text{asinh}(q/c)$ terms.
4. **The $1/(p + 1) - 1/(p + 2)$ cancellation** in the polynomial moment, worth 1.19 digits at $m = 9$ (Section 5.5).

5. **Do not recover a centred moment by subtraction.** Computing $J_{\text{fal}} = h\zeta + C$ and then recovering C as $J_{\text{fal}} - h\zeta$ destroys it: the two terms agree to eight digits. C must be returned by its own formula. $6.57 \rightarrow 1.01$ digits.
6. **Nested atan differences have an order.** One of the two nesting orders always degenerates and they never degenerate together; selecting by the larger relative separation took $11.78 \rightarrow 0.62$ digits.
7. **Float64 facePair coordinates carry ulps.** $3s/2$ is not representable, so a `Float64` panel differs from the exact one by an ulp; this is visible as up to 0.11 digits of apparent loss, and as sliver boxes (Section 5.6).
8. **The digits-lost metric must convert its inputs exactly,** `BigFloat(a64)` and not `BigFloat(10)^i`.
9. **Missing even- m base cases.** Two of the discovery route's recursions had only the $m = -1$ base, so an even m recursed forever; found when the cross-check first ran at even m .
10. **A silent truncation cap.** The thin-axis expansion stopped at level 6 as well as on its tolerance; invisible in `Float64` (the tolerance fires at level 3) but capping `BigFloat` accuracy at about $(\text{thin ratio})^7$, which showed as 10^{-41} entries in a table whose neighbours were at 10^{-60} .
11. **Double-factorial ratios as one conversion.** `T(a // b)` rounds once; `T(a) / b` rounds twice and can promote a `Float32` through `BigInt`.
12. **Prescale by powers of two.** Without it, intermediates overflow `Float32` above a coordinate of about 250 at $m = 12$ and `Float64` above 10^{25} ; with it, the computation is exactly equivariant, and 62 premature NaNs became zero.

6 Verification

Everything in this section is reproduced by `verify.jl`, which writes its five tables into `notes/moments/tables/` as `a_correctness.txt`, `b_faces.txt`, `c_crosschecks.txt`, `d_digits.txt` and `e_timing.txt`.

6.1 Against the reference at the eleven prescribed shapes

The reference is `pairMom` of `notes/gen/ref/mom.jl` at 320 bits with orders $(44, 30, 44)$, cached in `refcache.txt`, for $m = -1$ and odd m ; for even m the reference is an exact `Rational{BigInt}` evaluation of the multinomial expansion of r^{2k} over the four-dimensional product domain, which is independent of the box decomposition, of both integrators and of `pairMom` alike. The shapes are the cube, the six slender ones at $10^{\mp 3}$ on each axis, $(1/32)^3$, and the three permutations of $(1/32, 1/32, 1/512)$.

code	cube	thinX	thinY	thinZ	fatX	fatY	fatZ	small	slv _{zzz}	slv _{zyz}	slv _{zzy}
S	6.5e-60	6.3e-44	6.3e-44	6.5e-60	6.3e-44	6.3e-44	6.5e-60	6.6e-60	6.6e-60	4.2e-44	4.2e-44
Ef	3.1e-60	3.8e-44	1.1e-47	3.1e-60	1.1e-47	3.8e-44	3.1e-60	5.3e-60	5.3e-60	2.4e-45	3.3e-44
Ec	2.8e-60	1.1e-45	2.6e-45	2.6e-45	7.4e-44	6.3e-44	6.3e-44	3.6e-60	7.5e-46	7.5e-46	2.6e-46
Vc	3.0e-60	1.1e-47	1.1e-47	3.0e-60	1.1e-47	1.1e-47	3.0e-60	3.8e-60	3.8e-60	2.4e-45	2.4e-45
Vp	3.7e-60	6.0e-48	1.1e-45	6.0e-48	3.4e-44	2.9e-47	3.4e-44	2.5e-60	1.3e-46	2.7e-46	1.3e-46
P1	2.2e-60	2.6e-60	2.6e-60	2.0e-60	2.9e-60	2.9e-60	4.5e-60	2.8e-60	7.2e-61	3.0e-60	3.0e-60
P2	1.9e-60	2.1e-60	2.1e-60	3.0e-60	3.8e-60	3.8e-60	3.3e-60	3.3e-60	3.1e-60	2.7e-60	2.7e-60
P1s	2.9e-60	2.7e-60	1.3e-60	3.7e-60	2.5e-60	3.8e-60	4.5e-60	2.9e-60	1.1e-60	1.6e-60	3.4e-60
P1s-b	2.6e-60	1.5e-60	1.5e-60	1.0e-60	3.4e-60	3.4e-60	4.9e-60	2.8e-60	3.8e-60	1.2e-60	1.2e-60
P2s	2.9e-60	3.3e-60	3.8e-60	1.9e-60	3.2e-60	2.4e-60	1.3e-60	4.0e-60	1.8e-60	3.8e-60	2.8e-60
P2s-b	3.1e-60	2.1e-60	2.1e-60	1.6e-60	4.5e-60	4.5e-60	3.5e-60	3.8e-60	1.7e-60	4.1e-60	4.1e-60
Xg	9.6e-61	3.3e-60	6.6e-61	2.5e-60	3.1e-60	3.8e-60	3.8e-60	2.2e-60	2.1e-60	9.0e-61	2.6e-60
Xg-b	2.2e-60	3.0e-60	3.1e-60	2.4e-60	2.6e-60	3.9e-60	1.8e-60	2.1e-60	4.4e-60	2.6e-60	2.4e-60
Xg-c	4.1e-60	1.9e-60	3.2e-60	3.2e-60	2.0e-60	2.5e-60	2.5e-60	2.5e-60	2.0e-60	2.0e-60	2.8e-60
Xg-d	1.9e-60	4.1e-60	2.6e-60	2.6e-60	4.8e-60	3.8e-60	3.8e-60	2.7e-60	2.3e-60	2.3e-60	3.1e-60

Worst relative error over $m = -1, \dots, 12$ per (sub-code, shape). Global worst 7.35×10^{-44} ; **zero of 2310 values above 10^{-30}** .

Two things in that table are worth pointing out. First, the even- m columns of the per- m breakdown land at 10^{-96} to 10^{-97} , which is exactly the 320-bit unit roundoff: the box decomposition and both integrators are *exact* there, and that is the sharpest statement available anywhere in this document, because the even- m reference is an exact rational number. Second, the 10^{-44} to 10^{-47} entries occur only for the five touching codes and only at 1000:1 aspect ratios, and they are the reference’s own refinement residual. That was established by refinement, not asserted: recomputing `pairMom` at three increasing order sets at each geometry’s worst shape gives

code	worst shape	orders (28, 16, 22)	(36, 22, 30)	(44, 30, 44)
S	$a = 10^3$	1.26e-22	2.27e-30	6.34e-44
Ef	$a = 10^{-3}$	7.60e-23	1.37e-30	3.82e-44
Ec	$a = 10^3$	1.46e-22	2.64e-30	7.35e-44
Vc	(512, 32, 32)	7.45e-24	5.62e-32	2.43e-45
Vp	$c = 10^3$	6.79e-23	1.22e-30	3.41e-44
P1	$a = 10^3$	7.36e-44	3.87e-56	1.81e-68
P2	$b = 10^3$	5.78e-44	3.09e-56	1.33e-68
P1s	$b = 10^3$	7.40e-44	3.65e-56	1.73e-68
P2s	$b = 10^3$	5.66e-44	2.97e-56	1.24e-68
Xg	$a = 10^3$	6.60e-44	3.19e-56	1.60e-68

Every row falls by 8 to 14 orders per order-step and shows no floor, which is the signature of an exact formula measured against a converging quadrature. *The verification is limited by the reference, not by the formulas*, and the honest statement of the correctness bar is therefore “at the reference’s own convergence floor”, which on the non-touching geometries is 10^{-68} and on the touching ones 10^{-44} .

6.2 All 144 face pairs

`faceMoments` against `pairMom` at $s = (3/7, 5/11, 2/3)$, 256 bits, $m = -1, \dots, 8$: 1440 values, **zero above 10^{-30}** , worst 1.91×10^{-48} at $\Delta = (0, 0, 0)$, xzL-xzL, code S, $m = -1$, which is `pairMom`’s Duffy floor. Per sub-code:

S	1.91e-48	Ec	4.84e-49	Vp	2.57e-49	P2	9.31e-76	P1s-b	6.42e-76
Ef	1.44e-48	Vc	1.37e-48	P1	1.12e-75	P1s	9.44e-76	P2s	4.72e-76
P2s-b	5.97e-76	Xg	1.24e-74	Xg-b	2.09e-74	Xg-c	5.77e-75	Xg-d	5.26e-75

6.3 Cross-checks, homogeneity, symmetries, orientation

Against the plain explicit closed forms of Sections 3 and 4, over the eleven shapes and $m = -1, \dots, 12$ at 320 bits: the coplanar and parallel codes agree with the polar forms at 10^{-83} to 10^{-90} and the perpendicular ones with the master formula at 10^{-95} and with the discovery forms at 10^{-90} to 10^{-91} . The 10^{-83} to 10^{-85} entries are all at fatZ (1000:1 offset) and are the explicit forms' own cancellation, not the box path's — which is the point of the box path.

Homogeneity $I_m(\lambda \mathbf{s}) = \lambda^{m+4} I_m(\mathbf{s})$ at $\lambda = 3$ and $1/7$: worst 2.09×10^{-95} . Orientation invariance under all six axis permutations of both panels: worst 2.89×10^{-96} . In the earlier development the sweep was even wider — 24 orientations per geometry (six permutations times four reflection states), worst 3.75×10^{-90} , and in the audit all 48 signed axis permutations of all 144 face pairs, worst 1.71×10^{-96} with zero of 69 120 values above 10^{-60} , plus the $A \leftrightarrow B$ swap at 9.16×10^{-97} .

The symmetry sweep is a two-sided test and is reported as such: a swap is a symmetry when its entry sits at the reference floor, and its absence shows as an order-unity number, the two being ninety-five orders apart so that nothing is ambiguous.

code	$a \leftrightarrow b$	$a \leftrightarrow c$	$b \leftrightarrow c$	code	$a \leftrightarrow b$	$a \leftrightarrow c$	$b \leftrightarrow c$
S	0	n/a	n/a	P2s	8.4e-01	1.8e-01	4.6e+01
Ef	1.5e+00	n/a	n/a	P2s-b	8.7e-97	1.8e-01	3.4e+01
Ec	8.4e-01	1.6e-01	9.1e-97	Xg	4.7e+02	1.5e-01	8.6e+02
Vc	8.3e-97	n/a	n/a	Xg-b	8.7e-01	4.9e-01	1.9e+01
Vp	7.3e+02	9.0e-97	3.7e+02	Xg-c	5.6e+00	9.7e-01	9.1e-97
P1	1.3e-96	2.7e-01	4.9e+01	Xg-d	8.4e-01	1.0e-01	8.1e-97
P2	1.3e-96	9.1e-01	7.0e+01	P1s	9.9e-01	4.2e-01	3.1e+01
P1s-b	1.3e-96	4.1e-01	5.4e+01				

The symmetries the brief named are all present: Ec in $b \leftrightarrow c$, Vp in $a \leftrightarrow c$, and S, Vc, P1, P2 in $a \leftrightarrow b$. The sweep found three more that follow from the same reasoning and were not on the list: P1s-b and P2s-b in $a \leftrightarrow b$ (both in-plane axes are shifted, symmetrically) and Xg-c, Xg-d in $b \leftrightarrow c$ (both indicator axes are a full cell away). Every other swap is absent, including P1s and P2s in $a \leftrightarrow b$, which is the discriminating check that the shifted codes are not being confused with their unshifted parents. The n/a entries are honest: `canonPanels` for S, Ef and Vc does not depend on c at all, so a swap involving c tests nothing, and reporting a number there would be vacuous.

6.4 Digits lost in Float64

The grid is $b = 1$, a and c over $10^{-6}, \dots, 10^6$, with $c = 0$ as well for the nine two-dimensional codes ($13 \times 14 = 182$ cells there, $13 \times 13 = 169$ for the six three-dimensional ones), and $m = -1, \dots, 12$: about 37 000 values.

code	-1	0	1	2	3	4	5	6	7	8	9	10	11	12
S	0.30	0.00	0.07	0.00	0.00	0.00	0.29	0.00	0.30	0.00	0.30	0.05	0.40	0.39
Ef	0.05	0.00	0.00	0.05	0.24	0.03	0.03	0.14	0.17	0.09	0.21	0.34	0.37	0.26
Ec	0.17	0.01	0.15	0.41	0.30	0.28	0.43	0.46	0.49	0.54	0.72	0.67	0.74	0.77
Vc	0.08	0.00	0.04	0.00	0.17	0.17	0.00	0.17	0.26	0.05	0.13	0.23	0.19	0.23
Vp	0.46	0.00	0.21	0.06	0.23	0.18	0.35	0.23	0.49	0.42	0.50	0.47	0.61	0.73
P1	0.48	0.00	0.29	0.12	0.41	0.24	0.36	0.45	0.42	0.44	0.46	0.53	0.60	0.53
P2	0.62	0.00	0.20	0.23	0.19	0.24	0.29	0.39	0.37	0.55	0.47	0.55	0.60	0.59
P1s	0.64	0.00	0.24	0.17	0.32	0.27	0.42	0.43	0.42	0.54	0.53	0.54	0.54	0.62
P1s-b	0.34	0.00	0.17	0.14	0.14	0.29	0.31	0.44	0.45	0.34	0.43	0.51	0.60	0.55
P2s	0.63	0.00	0.17	0.25	0.32	0.25	0.41	0.41	0.54	0.55	0.54	0.54	0.61	0.63
P2s-b	0.28	0.00	0.13	0.13	0.14	0.18	0.37	0.38	0.38	0.54	0.50	0.58	0.58	0.60
Xg	0.45	0.00	0.35	0.36	0.39	0.46	0.47	0.54	0.59	0.59	0.64	0.65	0.61	0.73
Xg-b	0.78	0.00	0.38	0.35	0.44	0.44	0.57	0.54	0.60	0.59	0.67	0.63	0.75	0.77
Xg-c	0.90	0.00	0.37	0.38	0.50	0.54	0.59	0.74	0.66	0.71	0.74	0.76	0.91	0.78
Xg-d	0.85	0.00	0.14	0.31	0.18	0.21	0.37	0.38	0.43	0.43	0.59	0.56	0.68	0.66

Global worst 0.91 digits, zero cells above 1.0. The per-code worst cells are listed in Sections 3 and 4; the full 15 grids are in `tables/d.digits.txt`. For scale, the same quantities for the plain closed forms range from 11.7 (Vc) through 12.8 (Xg), 16.1 (Ec) and 22.6 (Vp) to 29.3 (P2s), and the original Gila formulas that this work replaces lost up to 18.8 digits at $m = -1$ alone.

An independent version of the same measurement, run in the audit over all 144 face pairs at nine cell shapes including $2^{\pm 20}$ on each axis — that is, real face pairs at extreme cell aspect ratios rather than canonical panels — gives global worst 0.90 digits and zero of 18 144 values above the bar.

6.5 Genericity and cost

- `Float32` in gives `Float32` out with no type leak; worst relative difference from the `BigFloat` evaluation $1.12 \times 10^{-6} = 9.4 \epsilon_{32}$, i.e. 0.97 digits, so the digits bar holds in single precision too. After the prescaling fix, of 2310 (code, shape, m) triples 51 return `Inf`, all of them true overflows of `Float32`, and none returns `NaN`.
- `Rational{Int}` panels give *exactly* the values of the corresponding float panels, relative difference 0.0, for $T = \text{Float64}$ and `BigFloat`.
- `BigFloat` at 128 bits against 320 bits: worst 7.95×10^{-39} , i.e. 2.7×2^{-128} , so every $\epsilon(T)$ -based termination test tracks the working precision. A single `Float64` constant anywhere in the file would floor this at 10^{-16} ; a grep of every transcendental call and every numeric literal confirms there is none.
- Cost, `Float64`, all fourteen moments of one face pair: $87 \mu\text{s}$ (Ec, two boxes) to $973 \mu\text{s}$ (P1, four boxes). At 36 face pairs per cell pair that is 3 to 35 ms per contact pair, against `pairMom`'s few seconds per pair at 256 bits. The two-dimensional codes are the expensive ones because they pay for the centred series at every rung; that is a deliberate trade of time for digits.
- `verify.jl` in full: (a) 7.7 s, (b) 5 min 16 s, (c) 37.6 s, (d) 58.5 s, (e) 5.0 s; 7 minutes 5 seconds single-threaded.

6.6 The adversarial audit

An independent sub-agent was given the finished file and told to break it. It found **no wrong value**. In `BigFloat`, every geometry it could construct — on lattice, off lattice, plateau trapezoids, intersecting panels, degenerate panels, aspect ratios 10^{-6} to 10^6 , m up to 30, all 48 signed axis permutations — agrees with `pairMom` at the reference's own convergence floor. Specifically:

- **Off-lattice geometry:** 40 random axis-aligned pairs with exact dyadic endpoints (coplanar, parallel at random offsets, perpendicular, partial overlaps, difference boxes straddling the origin, one panel $30\times$ smaller), plus nine targeted cases including three-piece plateau trapezoids on both axes and intersecting perpendicular panels: worst 3.28×10^{-43} , at a touching configuration, i.e. the Duffy floor again. The plateau path, which the lattice never exercises, is correct.
- **Extreme aspect ratios:** shapes with $10^{\pm 6}$ on each axis, 320 bits, against `pairMom` at raised orders (60, 40, 60): worst 1.56×10^{-60} , while the reference's own order-44 to order-60 gap at the same entry is 8.09×10^{-45} . The reference limits the agreement by fifteen orders of magnitude at every one of the 76 (code, shape) pairs measured.
- **Higher m :** m up to 30 for all fifteen sub-codes, worst 8.93×10^{-61} against a reference gap of 7.51×10^{-46} ; no NaN or Inf at any m in `BigFloat`.
- **Branches and signs:** every helper containing an `atan`, `log`, `log1p` or `asinh` branch evaluated at 320 bits against an exact 1024-bit reference, on sweeps crossing each threshold from both sides to within 10^{-14} : worst 2.5×10^{-88} , most at 10^{-96} . At box level, each shape branch against the unbranched divergence path at 1024 bits, sweeping to within 10^{-9} of each gate: 9.7×10^{-211} (dominant-axis gate), 5.4×10^{-158} (thin gate), 4.7×10^{-294} (series gate). No jump at any of them.
- **The series indicator** is an honest upper bound in all 240 rows of its sweep, over-stating by a factor 3.4 or more and never under-stating.

The audit's five substantive findings were all about conditioning, precision ceilings or coverage claims, and all five were fixed: the off-lattice thin-indicator loss and the `BigFloat` ceiling of the sliver branch (Section 5.7), the premature `Float32` overflow (prescaling), and the unreachable `tol` keyword. The fifth, the $m > 12$ digit growth, was measured, its mechanism identified and documented, and deliberately not fixed (Section 5.8). The fix pass was itself regression controlled: all 39 438 values of the digits grid and all eleven reference shapes are *bitwise identical* before and after, and `tables/a_correctness.txt` is byte-identical to the copy taken before the work.

7 Comparison with Gila's numerics

This is the end-to-end test: it exercises the geometry bookkeeping as well as the algebra, because a formula that is right but attached to the wrong face pair fails it. All of it is in `tables/f_gila_{a,b,c}.txt`.

7.1 The entry-to-geometry map

The map from Gila's `wS`, `wE`, `wV` entries to the sub-codes was read out of `wekGrdPts!` and `wekEDir/wekVDir` and then confirmed numerically. In the local frame $(g_X, g_Y, g_Z) = \text{grdScl}(\text{dir}, \text{s})$, panel A is $[0, g_X] \times [0, g_Y] \times \{0\}$ and the second panel of each call identifies the geometry: `wS` is `S`, `wE` entries 1–6 are `Ef`, entries 7–9 are `Ec` (each averaged over two orientations), `wV` entries 1–3 are `Vc` and 4–6 are `Vp` (again averaged over two). The confirmation is that I_{-1} from the closed forms equals $4\pi f^2$ times Gila's own static part where Gila has one: 2.51×10^{-16} for `rSrfS1f`, 1.02×10^{-16} for `rSrfEdgFlt`, 2.54×10^{-16} for `rSrfEdgCrn`, at every cell scale. And the two orientations that Gila averages give *bit-identical* closed-form moments, relative difference exactly zero at all four scales: `Ec` by its $b \leftrightarrow c$ symmetry, `Vp` by its $a \leftrightarrow c$ symmetry. Gila's $(\text{xyA} + \text{xyB})/2$ is an average of two evaluations of the same integral.

7.2 The weak integrals: wekS, wekE, wekV

The series $\frac{1}{4\pi f^2} \sum_n \frac{(2\pi i f)^n}{n!} I_{n-1}$, summed in 256-bit arithmetic from closed-form moments only, against Gila’s `intOrd` 48 and 64 values at four cell scales and two frequencies:

code	<code>intOrd</code> 48	<code>intOrd</code> 64
S	6.30e-11	1.37e-11
Ef	2.90e-10	1.92e-11
Ec	9.29e-11	6.06e-12
Vc	4.98e-10	8.99e-11
Vp	1.80e-09	3.26e-10

Worst relative difference per code; the order-64 column is over the three cell scales that carry both orders. These reproduce the error estimates of Section 5.4 of the companion document quantitatively: 6×10^{-11} for self and edge at $\lambda/4$, 9×10^{-13} at $\lambda/32$, 7×10^{-11} for the vertex integrals at all scales, and 1.8×10^{-9} for the vertex integrals on the slender cell.

The direction of the discrepancy is the point. Of the 108 rows carrying both orders, 99 converge (the order-64 value is closer to the series in both components), and 21 of those bracket it (the two Gila values straddle the series in the real part, so the series lies between them). *Gila’s own numerics converge toward the closed-form series*, which is the strongest available confirmation that the new formulas and the existing code compute the same integral.

Nine rows fail that criterion, and seven of them fail only as an artefact of the test: their imaginary parts are already at round-off and cannot improve, so a criterion demanding improvement in *both* components marks them as plateaus even though the real part improves (by a factor of four for three of them, and slightly for the other four). The two genuine failures are the documented one. On the slender cell (1/32, 1/32, 1/512) the flat-edge entry **E2** differs from the series by 3.105×10^{-13} at order 48 and by 3.233×10^{-13} at order 64: refinement moves *away* from the truth, while the 48-to-64 spread is only 1.28×10^{-14} . Section 5.4 of the companion document flagged exactly this (“differs from the series by 3×10^{-13} while its own spread is 1.3×10^{-14} ”), and it is reproduced here to two digits at both frequencies. Refinement-based convergence tests are not a reliable error proxy for slender cells.

Two further facts. A `Float64` evaluation of the series with `Float64` moments agrees with the 256-bit sum to 9.12×10^{-16} over all rows, about 4 units in the last place, so nothing is lost by summing in double precision at these cell sizes. And the term counts needed for 10^{-16} relative accuracy are 10–13 at $\lambda/32$, 15–21 at $\lambda/8$, 20–29 at $\lambda/4$, identical at $f = 1$ and $f = 1 + 0.1i$: one more term than Section 5.4’s table, which used 10^{-15} .

Cost: the fourteen Gila runs behind that table took 2160 seconds single-threaded. One `Float64` series sum takes $0.42 \mu\text{s}$ once the moments are built, and the moments do not depend on f .

7.3 The order-9 fixed rule on the non-touching pairs

Gila evaluates the touching-shell pairs with a fixed order-9 tensor Gauss–Legendre rule. The raw face-pair integral is `srfMat[fp] × Vt`, which was verified by an independent `BigFloat` re-implementation of the same rule agreeing with Gila’s value to 2.3×10^{-16} – 1.2×10^{-14} . The comparison covers all 70 non-touching pairs of the four offsets (6 + 15 + 22 + 27, matching the counts of Section 3.2 of the companion document) at four cell scales.

code	$(1/32)^3$	$(1/8)^3$	$(1/32, 1/32, 1/512)$	$(1/4)^3$
P1	8.92e-14	9.27e-14	1.57e-02	9.46e-14
P2	1.10e-15	1.75e-15	6.16e-15	9.17e-16
P1s	2.12e-14	2.33e-14	2.81e-04	2.28e-14
P1s-b	9.92e-15	9.58e-15	1.80e-07	7.03e-15
P2s	5.71e-15	5.32e-15	4.27e-15	4.03e-15
P2s-b	1.90e-15	6.31e-15	1.28e-08	6.80e-15
Xg	1.45e-14	1.43e-14	8.30e-15	1.72e-14
Xg-b	4.09e-15	3.76e-15	1.37e-06	6.74e-15
Xg-c	2.74e-15	1.41e-15	5.66e-15	7.25e-15
Xg-d	7.32e-15	3.56e-15	7.20e-15	7.59e-15

Worst relative difference between Gila's order-9 value and the series, per (sub-code, cell scale), at $f = 1$.

On cubic cells the rule is better than expected: 3.5×10^{-16} to 1.1×10^{-13} across every pair, scale and frequency, with the worst being the P1 pair (faces one cell apart, where the kernel peaks hardest) and the best the two-cell P2 and P2s pairs at 7×10^{-16} . That is consistent with the code comment behind `cntOrd = 9`.

On the slender cell $(1/32, 1/32, 1/512)$ it breaks down, for exactly the pairs whose *gap* is the short edge. Thirteen of the 70 pairs exceed 10^{-9} relative, the worst being 1.57×10^{-2} for the P1 pair $\Delta = (0, 0, 0)$, xyL-xyU (two $1/32 \times 1/32$ panels $1/512$ apart), then 2.81×10^{-4} for the laterally shifted version of the same pair, 1.37×10^{-6} for the Xg-b pairs whose perpendicular gap is $1/512$, and 1.80×10^{-7} and 1.28×10^{-8} for the doubly shifted pairs. Gila matches an independent implementation of its own rule to 10^{-14} on those rows, so this is the rule's truncation error and not a mapping error. With the assembly amplification of 10 to 1000 documented in Section 3.3 of the companion document, 1.6×10^{-2} at the face-pair level is not a small number. `cntOrd = 9` was measured on cubic cells only; it is not safe for slender cells. This is a finding about the existing code rather than about the present work, and the series removes it.

One bookkeeping correction fell out of the same exercise: at $\Delta = (1, 1, 1)$ the pairs xyL-xyU and yzL-yzU are Vc, not P1s (the two faces become coplanar at that offset), and $\Delta = (1, 0, 0)$ yzL-yzU is S, not P1s. The genuine doubly shifted P1s at $\Delta = (1, 1, 1)$ is yzL-yzL. Applying the order-9 rule to the mis-identified pairs gives 8.3×10^{-7} relative error, and applying it to the code-S pair returns Inf + NaNi — which is correct behaviour, since `egoSrfFxd!` is documented as valid only for pairs that are not in contact and the production path never calls it on them.

7.4 Frequency homogeneity

Every integral of $1/r$ is proportional to $1/f^2$ and nothing else, and term n of the series scales exactly as f^{n-2} . Checked at $f = 1, 1 + 0.1i$ and 0.37 on a $(1/8)^3$ cell for S, Ef, Ec, Vc, Vp and a P1 shell pair: f^{2-n} term _{n} agrees between frequencies to between 0 and 4.7×10^{-77} for $n = 0, \dots, 9$, i.e. at the 256-bit round-off. On the Gila side, f^2 times `rSrfS1f`, `rSrfEdgFlt` and `rSrfEdgCrn` agrees with $I_{-1}/(4\pi)$ to 2.5×10^{-16} or better at all three frequencies. This is the one-line test that would have caught the missing $1/f^2$ of Section 4.5 of the companion document, and the closed forms now pass it by construction.

8 The registry of approach families

The search was run as a portfolio: several mathematical mechanisms alive at once, kept independent in the early rounds so that they would not all converge on the same attractive reduction, with a

written registry of what each had achieved and where it was blocked. That discipline is the reason the result is believable, and the failures are as instructive as the successes, so both are recorded.

8.1 (a) Polar and Duffy coordinates with the secant-power reduction

Mechanism. On a two-dimensional box with a corner at the origin, polar coordinates make the radial integral of r^m exact — it is a power of the boundary radius $p \sec \theta$ or $q \csc \theta$ — and the angular integrals are of $\sec^j \theta$ and $\csc^j \theta$, which reduce by $\int \sec^j = \frac{\sec^{j-2} \tan}{j-1} + \frac{j-2}{j-1} \int \sec^{j-2}$ down to $\int \sec = \operatorname{asinh} \tan$ and $\int \sec^2 = \tan$. This is Section 5.2 of the companion document generalized to trapezoidal weights and to a frozen offset c .

Outcome: complete for S, Ef, Vc, P1, P2, P1s, P2s, all integer $m \geq -1$, at the reference floor, and it produced the assembly (15) and the corner moments G_{ij} of (17). The double-factorial coefficients of (5) appear here as the sec-reduction coefficients, and the identification $J_K(L, g) = g^{K+1} S_{K+2}(L/g)$ is what showed that this route and the divergence route compute the same object.

Where it stopped. It was not attempted on Ec, Vp or Xg, because a perpendicular pair does not reduce to a two-dimensional difference variable and the polar reduction has nothing to act on. After regrouping it reached 1.3 (S) to 2.0 (Vc) digits lost, above the bar; the residual weaknesses it named — the four-quadrant inclusion–exclusion of Vc, and the weighted primitives it lacked — are exactly what the box route later supplied.

8.2 (b, d) The divergence recursion as an algorithm

Mechanism. Theorem 5.1’s proof is constructive, so implement it: represent the domain as the product box, apply (3), restrict on each facet, drop null directions, complete the square to find the new offset, recurse, and integrate the one-dimensional terminal cases in closed form. Families (b) and (d) of the original plan — the divergence recursion and m -recurrences from $\nabla \cdot (Xr^m) = (N+m)r^m$ — are the same identity and were merged.

Outcome: complete for all ten Table-1 codes, all m from -1 to 12 , agreeing with pairMom between 10^{-36} and 10^{-85} , which order refinement showed to be the reference’s residual. Every identity in the architecture was checked numerically before use and none was wrong: the M_{11} recurrence to 3.6×10^{-81} , the unrolled J against its recurrence to 5.7×10^{-96} , and the collapse $K^{(-1)} = N_0$ to 1.8×10^{-96} . This route is the one that generalizes without thought to any axis-aligned pair, and it is the skeleton of the shipped file.

What it left. As written it cancels badly: 28 to 29 digits for the P family (all of it inside M_{11}), 16 for Ec and Xg (inside the K_u difference), 12 to 13 for S, Ef and Vc (inclusion–exclusion at extreme aspect ratio). Its diagnosis named the exact sub-expressions, which is what made the cancellation rounds tractable.

8.3 (c) The box potential and integration by parts

Mechanism. Reduce a perpendicular pair to $\int (\text{weight}) \Phi_m(u) du$ with Φ_m the potential of a rectangle at a point, then integrate along the shared direction, as **rSrfEdgCrn** was originally derived.

Outcome: complete for Ec, Vp and Xg for all integer m , and it produced the master formula (28) covering every perpendicular face pair of Table 1 including the Xg variants: 960 values at 1.53×10^{-51} . Its central discovery is (12): because $xr^m dx$ is an exact differential, the by-parts integral is $TK^{(m)}$ minus a plain difference of Ψ at exponent $m+2$, and *the antiderivative of an arctangent, which the naive by-parts route seems to demand, is never needed*. That one identity is why the perpendicular geometries turned out to be no harder than the coplanar ones.

Where it stopped. Its arrangement is a corner-box form plus inclusion–exclusion, which regroups to 1.2 to 2.1 digits — above the bar, and above it for every geometry at high m (its measured range on the aspect grid is 0.15 to 1.86). It correctly identified the fix as an offset-range primitive obtained by applying the divergence identity on the shifted box, and did not implement it; that became the shifted-box development of Section 5.5.

8.4 (e, f) Numerical discovery and symbolic integration

Mechanism. Compute a moment to 200 bits at rational geometry, conjecture a basis of elementary atoms (powers of the lengths, hypotenuses, asinh and atan of length ratios), recover the rational coefficients by solving a linear system, then derive them. SymPy on the reduced one- and two-dimensional integrals supplied the conjectures for fixed m , and the pattern across m supplied the general law.

Outcome: complete for Vc, Vp and Xg for all m , with explicit forms at $m = -1, 1, 3, 5$ (Vc also at 7) and *all 259 rational coefficients recovered independently from pairMom data*, residuals 3.9×10^{-30} to 1.8×10^{-53} . It produced the general- m Vc formula (22) with its universal hypotenuse polynomial $\hat{G}_m$, and the atom-coefficient laws for Vp and Xg — the observation that the atom set never changes with m (six atoms for Vc, fourteen for Vp, sixteen for Xg) and only the coefficients move, homogeneously of degree $m + 3$. It also found, from the algebra rather than from the classification, that one Xg variant has exactly Vp’s reduced integral.

Where it stopped. Cancellation was not addressed at all: up to 22.6 digits for Vp. And it explicitly noted that Xg in Table 1 covers at least three distinct reduced integrals without enumerating them, which is the same discovery the signature classification made from the other direction.

8.5 (g) Generating functions: deprioritized, and why

Mechanism. Sum the moments against t^m to a closed form in t and read the moments off, or relate the moment sequence to the special-function form of Section 5.3 of the companion document.

Outcome: not pursued, on a specific ground rather than on effort. The exponential generating function of the moments *is* the face-pair integral of the full kernel, $\sum_m t^m I_m / m! = \iint e^{tr}$, and Section 5.3 of the companion document establishes that this object is not elementary: for coplanar pairs it reduces to incomplete Hankel functions of the family F_m , which have no certified evaluator and whose small-argument evaluation is the Taylor series in disguise, and for perpendicular pairs no reduction to one-variable special functions was found at all. So a closed form for the generating function is a closed form for the thing we are trying to expand, and reading the moments off it is re-expanding the series — the moments are the coefficients, and one computes them directly. The route was registered as reopenable only if a sub-agent proposed a mechanism that avoids the full kernel; none did, and none was invented.

This is worth stating plainly because it is the one place where a natural-looking approach is not merely hard but circular.

8.6 The claimed 1.16-digit floor, and its resolution

The most instructive episode of the whole search. The second round measured the doubly shifted two-dimensional seed and reported a floor: every exact elementary form of $W(\text{fall}, \text{fall})$ has the four coefficients $u_2 v_2$, $-u_2$, $-v_2$, 1 on the four functionally independent moments, so the amplification $u_2 v_2 / (h_u h_v) = 16$ is forced, and the measured assembly-only loss with all primitives exact was 1.16

digits, matching $\log_{10} 16 = 1.20$. The conclusion drawn was that 1.16 could not be beaten without double-double arithmetic.

The reasoning is correct and the conclusion is wrong, and the gap between them is the lesson. The floor is a statement about *closed forms*, not about the integral: the centred decomposition has amplification exactly 1.0 with exact centred moments, and the centred moments are perfectly well defined objects. What the floor really says is that their elementary antiderivatives cancel intrinsically (Section 5.3) — which was then measured, at 1.2×10^{25} for X and 4.8×10^{37} for Z , and is a much sharper and more useful statement than the original claim. The resolution was to compute the centred moments by an exact finitely truncated expansion instead (Section 5.4), which took the worst box type from 1.83 to 0.77 digits and every geometry under 0.87.

The procedural lesson: a floor argument should be tested by measuring the quantity it claims to bound, in the arrangement it claims is optimal, before it is accepted. Here the measurement cost an afternoon and moved the deliverable from “one and a bit digits” to “under one digit everywhere”.

8.7 The atan antiderivative that was never needed

Worth its own note as a near-miss. Integrating the box potential N_0 along the shared direction, as `rSrfEdgCrn` was derived, appears to require the antiderivative of $u^2 \operatorname{atan}(bc/(uR))$ and its relatives — messy objects that would have to be found for every m . The identity (12) makes them unnecessary: the x -weighted part of the integral is exactly $\Psi_{m+2}(T) - \Psi_{m+2}(0)$, because $x r^m dx$ integrates by inspection, and the unweighted part is the box moment $K^{(m)}$, which has its own recurrence. Two of the four families spent time on the antiderivative before this was noticed. It is the single most useful simplification in the perpendicular half of the problem.

8.8 Round structure

For the record, the search ran in four rounds. Round one launched four derivation families in parallel plus an infrastructure agent (geometry catalogue, reference cache); it returned all ten geometries solved by at least two independent routes, plus the finding that Table 1 lumps inequivalent subgeometries, which forced the fine classification. Round two attacked cancellation in two and three dimensions separately and ran the Gila comparison; it returned the three-dimensional bar met, the two-dimensional bar missed with a floor argument, and the sliver-box bug. Round three resolved the floor argument and integrated everything into one file with its verification. Round four was an adversarial audit by an agent that had not seen the development, followed by a fix pass under bitwise regression control.

9 What remains uncertain or unfinished

The verification is limited by the reference, not by the formulas. Every correctness number in Section 6 is `pairMom`’s own refinement residual. Raising the reference’s orders drops the disagreement by 8 to 14 orders per step with no floor, at every geometry and every shape tested, so what has actually been demonstrated is agreement to 10^{-44} (touching pairs) or 10^{-68} (non-touching pairs), and the true accuracy of the closed forms in extended precision is unknown from below. The even- m columns are the exception and the reassurance: there the reference is an exact rational number, and the agreement is at the 320-bit unit roundoff, 10^{-96} , meaning the box decomposition and both integrators are exact.

Two evaluation schemes are not closed forms, and are labelled as such. The centred Taylor branch (Section 5.4) is an exact, finitely truncated expansion with elementary coefficients and exact weight moments, used only where the elementary antiderivatives are ill-conditioned; it is not a quadrature, but neither is it a closed form, and the closed form it is evaluating is the one in Section 3. The off-lattice thin-indicator branch (Section 5.7) genuinely *is* a quadrature — a precision-adaptive Gauss–Legendre rule along the thin axis — and it is the one place in the file where that word applies. It is never reached for the fifteen lattice sub-geometries, and the evidence is bitwise: all 39 438 values of the digits grid are identical before and after it was added. A reader who needs the deliverable to be closed-form all the way down should note that the general entry point `pairMoments(A, B, mMax)` accepts arbitrary axis-aligned pairs, and for a sufficiently thin off-lattice indicator axis it will take that branch.

Digits above $m = 12$. The one-digit property holds for $m \leq 12$ and degrades predictably above it, $\log_{10}(\text{rungs}) + 0.08$ with $\text{rungs} = (m + 1)/2$: 1.15 digits for Ec and 1.11 for Xg-c at $m = 25$, 1.28 for Ec at $m = 30$, with 11 cells above one digit out of 1785 measured on a coarse grid — and the published 13×13 grid would probably find slightly worse. Since $m = 25$ is what a $\lambda/4$ cell needs, this is a real limitation for coarse cells, not a curiosity. The cause is identified (one rounding per rung of the three-dimensional one-dimensional ladder, no cancellation involved) and the two fixes are known (an m -independent evaluation of the three-dimensional primitives for the general box shape, or a compensated ladder). Neither was done, because both change every `Float64` value in the published tables and this pass was regression controlled. Correctness at high m is unaffected: 8.9×10^{-61} against `pairMom` at $m = 30$.

Float32 overflows. At fatX and fatZ shapes and $m \geq 10$, 51 of 2310 `Float32` results are Inf. All 51 are true overflows — the exact value exceeds `floatmax(Float32) =  $3.4 \times 10^{38}$` — and after the prescaling fix none is a NaN produced by an Inf – Inf inside the identities, which was the dangerous failure mode (it discarded 21 representable answers silently). The same mechanism sits at about 10^{307} in `Float64`, i.e. at a coordinate ratio around 10^{25} at $m = 12$.

The order-9 rule on slender cells. Section 7 documents relative errors up to 1.6×10^{-2} at the face-pair level for the non-touching pairs whose gap is the short edge of a 16:1 cell, against 10^{-14} on cubic cells. This is a finding about Gila’s current contact path and not about this work: `cntOrd = 9` was measured on cubic cells, and the measurement did not cover the case where the *gap* rather than the panel is the small length. With the assembly amplification of 10 to 1000 it is not negligible. The series removes it, which is one more reason to wire this in.

Coverage claims that were overstated and are now corrected. The canonical panels of S, Ef and Vc do not depend on c , so their “ 13×14 ” digits grids contain 13 distinct geometries, their eleven-shape rows contain eight distinct geometries, and the $a \leftrightarrow c$ and $b \leftrightarrow c$ symmetry entries for those three codes test nothing. The tables now say so and print `n/a` rather than a vacuous number. The numbers themselves were never wrong; only the counts were inflated.

Small things left as they are. Reversed intervals (a panel given with $x_1 > x_2$) return zero silently rather than raising, matching the reference `pairBox`’s behaviour; `momentSeries` at $f = 0$ returns NaN rather than erroring; and the thin-axis expansion has a relative floor around 10^{-158} at 1024 bits, invisible at 512 bits and below. None affects anything the deliverable claims.

Two sub-codes rest on numerics alone. P1s-b and P2s-b have no independent explicit closed form in any of the four families. Structurally they are the Vc assembly with a frozen offset, and the box route computes them; their evidence is the 320-bit reference cache, the 144-pair sweep, homogeneity, orientation invariance and the $a \leftrightarrow b$ symmetry that distinguishes them from P1s and P2s. Writing out the collected explicit form, which is mechanical, would close the gap.

The Gila comparison tables and the fix pass. The audit fixes changed Float64 face-pair values by at most 4 units in the last place (9.2×10^{-16} relative) on panels whose coordinates are inexact, which is what Gila’s cell coordinates are. The (f) tables were produced around that boundary. Nothing they report at the 10^{-15} level or coarser can move, but a last-digit agreement quoted from them should be regenerated before being relied on.

What is now settled. Theorem 5.1 of the companion document — that every I_m with integer $m \geq -1$ is elementary — was described there as “a well-founded plan rather than a finished proof until the first odd moments have been written out”. It is now confirmed constructively for all fifteen sub-geometries and every integer $m \geq -1$: the closed forms exist, they are built from polynomials, square roots of sums of squares, asinh and atan exactly as the theorem predicts, and they have been verified against an independent evaluator at the evaluator’s own floor. Nothing in the theorem’s proof sketch turned out to be wrong, and its exceptional cases ($N' + m = 0$) appear exactly where predicted, as the asinh and atan seeds.

The next task. Wiring this into Gila. It was deliberately out of scope, and Section 10 is the plan.

10 Implementation notes for Gila

Section 6.8 of the companion document sketched the integration point; here is the version informed by what the moments actually cost and by what the comparison found.

1. **Where. wekTrp**, which already memoizes the weak integrals per cell shape. The moments are frequency independent, so the memo key loses its f dependence entirely and a frequency sweep on a fixed geometry rebuilds the contact terms for nothing.
2. **What to compute once per cell shape.** For each of the fifteen sub-codes and the three face orientations, I_m for $m = -1, \dots, M$: a few hundred real numbers, at 87 to 973 μs per face pair for all fourteen moments. Against the 5 seconds per cubic cell shape and 20 to 35 seconds per slender shape that DIRECTFN currently costs, this is free.
3. **What to compute per frequency.** $\frac{1}{4\pi f^2} \sum_n \frac{(2\pi i f)^n}{n!} I_{n-1}$ in Float64: 0.42 μs per entry, 10 to 13 terms at $\lambda/32$ and 20 to 29 at $\lambda/4$. Summing to $M = 12$ covers cells up to about $\lambda/8$; $\lambda/4$ needs $M \approx 30$. Use the returned indicator $|\text{last term}|/|\text{sum}|$ as the gate: it is an honest upper bound on the truncation error, over-stating by a factor of at least 3.4 in every case measured.
4. **Keep the DIRECTFN fallback behind a cell-size gate.** The series loses about $kD/\ln 10$ digits for coarse cells; below $\lambda/4$ the loss is under one digit and the Float64 sum is within 4 units in the last place of a 256-bit one, so the gate can sit around $kD \approx 3$ as the companion document proposed.
5. **Assemble with the existing conventions:** raw integrals divided by the cell volume, and the Ec and Vp cross terms averaged over the two orientations — which become *exactly* equal, since the closed forms give bit-identical values for the two orientations. That average can therefore be dropped, halving that part of the work.

6. **Replace the order-9 rule for the touching-shell pairs with the same series.** All five shell codes and their four sub-codes are covered here, and the comparison shows the rule failing by 1.6×10^{-2} on slender cells, so this is a correctness fix and not only a speed one.
7. **Feed exact panel coordinates.** `facePair` computes a face coordinate as $\text{off} \cdot s + \text{sig} \cdot s/2$, and $3s/2$ is not representable in binary, so `Float64` cell coordinates produce panels an ulp away from the exact ones and hence ulp-wide sliver boxes. Gila stores `scl` as exact `Rationals` already, so the clean route is to hand `pairMoments` the rational panels; the `tol` keyword is the fallback if that is inconvenient.
8. **Test the way this document tests.** Compare each new closed form with the extended-precision evaluator on an aspect-ratio grid, check homogeneity in f (the one-line test that would have caught the historical missing $1/f^2$), and check the assembled operator against the current `DIRECTFN` values at two orders, which must bracket the new value — as they do here for 21 of the 108 rows and converge toward it for 99.

Acknowledgement of method

This work was organized by me as the root agent (Claude Fable 5.1). The derivations, the cancellation analyses, the numerical verification and the adversarial audit were carried out by sub-agents running Claude Opus, kept independent of one another in the early rounds so that the four derivation families would not converge prematurely; mechanical transcription, classification sweeps and tabulation were done by sub-agents running Claude Sonnet. I read only the sub-agents' written reports, never their transcripts, and required every report to contain actual formulas, actual numbers, and the exact point at which anything was stuck. Every number in this document is traceable to one of those reports or to a table in `notes/moments/tables/`. Paul Virally offered Mathematica access on request; it was not needed and was not used, and no symbolic algebra system beyond SymPy in a throwaway environment appears anywhere in the chain.

The synthesis, the architecture of Section 2, the decision to reject the 1.16-digit floor argument, and this document are mine.

A The sub-geometry map

The four tables below give the sub-code of every face pair (Δ, F, F') of the source cell and its three kinds of touching neighbour: rows are the target face F (on the cell displaced by Δ), columns the source face F' (on the cell at the origin), in Gila's face order `yzL yzU xzL xzU xyL xyU`. The other 22 touching neighbours are these up to reflection. Compared with Table 1 of the companion document, the entries `Xg`, `P1s` and `P2s` are refined into `Xg`, `Xg-b`, `Xg-c`, `Xg-d`, `P1s`, `P1s-b`, `P2s` and `P2s-b` by the exact per-axis convolution signature of Section 2.4; every other entry is unchanged. Each of the 144 assignments was confirmed numerically against the canonical panels of the named sub-code, 144/144 at a maximum relative error of 1.60×10^{-74} .

$\Delta = (0, 0, 0)$: the cell itself							$\Delta = (1, 0, 0)$: face neighbour						
$F \backslash F'$	yzL	yzU	xzL	xzU	xyL	xyU	$F \backslash F'$	yzL	yzU	xzL	xzU	xyL	xyU
yzL	S	P1	Ec	Ec	Ec	Ec	yzL	P1	S	Ec	Ec	Ec	Ec
yzU	P1	S	Ec	Ec	Ec	Ec	yzU	P2	P1	Xg	Xg	Xg	Xg
xzL	Ec	Ec	S	P1	Ec	Ec	xzL	Xg	Ec	Ef	P1s	Vp	Vp
xzU	Ec	Ec	P1	S	Ec	Ec	xzU	Xg	Ec	P1s	Ef	Vp	Vp
xyL	Ec	Ec	Ec	Ec	S	P1	xyL	Xg	Ec	Vp	Vp	Ef	P1s
xyU	Ec	Ec	Ec	Ec	P1	S	xyU	Xg	Ec	Vp	Vp	P1s	Ef

$\Delta = (1, 1, 0)$: edge neighbour						
$F \setminus F'$	yzL	yzU	xzL	xzU	xyL	xyU
yzL	P1s	Ef	Xg	Ec	Vp	Vp
yzU	P2s	P1s	Xg-c	Xg	Xg-b	Xg-b
xzL	Xg	Ec	P1s	Ef	Vp	Vp
xzU	Xg-c	Xg	P2s	P1s	Xg-b	Xg-b
xyL	Xg-b	Vp	Xg-b	Vp	Vc	P1s-b
xyU	Xg-b	Vp	Xg-b	Vp	P1s-b	Vc

$\Delta = (1, 1, 1)$: vertex neighbour						
$F \setminus F'$	yzL	yzU	xzL	xzU	xyL	xyU
yzL	P1s-b	Vc	Xg-b	Vp	Xg-b	Vp
yzU	P2s-b	P1s-b	Xg-d	Xg-b	Xg-d	Xg-b
xzL	Xg-b	Vp	P1s-b	Vc	Xg-b	Vp
xzU	Xg-d	Xg-b	P2s-b	P1s-b	Xg-d	Xg-b
xyL	Xg-b	Vp	Xg-b	Vp	P1s-b	Vc
xyU	Xg-d	Xg-b	Xg-d	Xg-b	P2s-b	P1s-b

Counts over the 144 pairs: Ec 34, Vp 22, Xg-b 20, Xg 12, P1 8, P1s 8, P1s-b 8, S 7, Ef 6, Xg-d 6, Vc 5, P2s-b 3, P2s 2, Xg-c 2, P2 1. The non-touching pairs (masked 0 in **egoFunSng!**, evaluated by the order-9 rule) number 6, 15, 22 and 27 for the four offsets, matching Section 3.2 of the companion document; the count of 70 over the four offsets is what Section 7 sweeps.

The canonical panels of the fifteen sub-codes, as **canonPanels** defines them with A or the first panel listed first, are

code	panels	code	panels
S	A, A	P2s	$A, [a, 2a] \times [0, b] \times \{2c\}$
Ef	$A, [a, 2a] \times [0, b] \times \{0\}$	Xg	$A, [0, a] \times \{2b\} \times [0, c]$
Ec	$A, [0, a] \times \{b\} \times [0, c]$	Xg-b	$\{2b\} \times [a, 2a] \times [0, c], [0, b] \times [0, a] \times \{0\}$
Vc	$A, [a, 2a] \times [b, 2b] \times \{0\}$	Xg-c	$\{2b\} \times [c, 2c] \times [0, a], [0, b] \times \{0\} \times [0, a]$
Vp	$A, \{a\} \times [b, 2b] \times [0, c]$	Xg-d	$\{2b\} \times [c, 2c] \times [a, 2a], [0, b] \times \{0\} \times [0, a]$
P1	$A, [0, a] \times [0, b] \times \{c\}$	P1s-b	$[a, 2a] \times [b, 2b] \times \{0\}, [0, a] \times [0, b] \times \{c\}$
P2	$A, [0, a] \times [0, b] \times \{2c\}$	P2s-b	$\{2c\} \times [a, 2a] \times [b, 2b], \{0\} \times [0, a] \times [0, b]$
P1s	$A, [a, 2a] \times [0, b] \times \{c\}$		

with $A = [0, a] \times [0, b] \times \{0\}$. Note that S, Ef and Vc do not use c at all, which is why the symmetry sweep reports **n/a** for swaps involving it.

B The files in notes/moments

file	what it is and what it produces
<code>moments.jl</code>	The deliverable. 1720 lines, no dependencies, generic in the number type. Sections in order: the two-dimensional box integrator (<code>asnD</code> , <code>dchs</code> , <code>ctrM1</code> , <code>jFal</code> , <code>lad1</code> , <code>offD</code> , <code>potCrn</code> , <code>ddAtn</code> , <code>potBox</code> , <code>box2Tay</code> , <code>box2lad</code> , <code>box2</code>); the three-dimensional box integrator (<code>jSeg</code> , <code>djSeg</code> , <code>wmSeg</code> , <code>wpSeg</code> , <code>slAng</code> , <code>psiBox</code> , <code>kBox3</code> , <code>wmBox3</code> , <code>wpBox3</code> , <code>polMom</code> , <code>box3Ser</code> , <code>box3Thn</code> , <code>box3Slv</code> , <code>box3</code>); the box decomposition with exact breakpoints (<code>axsCnv</code> , <code>pairBxs</code> , <code>pairBxsTol</code>); the geometry layer (<code>facePair</code> , <code>classifyFull</code> , <code>geomCode</code> , <code>canonPanels</code> , <code>loadRef</code>); the entry points (<code>pairMoments</code> , <code>faceMoments</code> , <code>momentSeries</code>); and the plain explicit closed forms, marked as cross-checks only.
<code>verify.jl</code>	The verification driver, <code>julia --startup-file=no verify.jl a b c d e</code> . Writes the five tables below. Total 7 min 05 s single-threaded.
<code>gilacmp.jl</code>	The Gila comparison, <code>julia --startup-file=no --project=<env> gilacmp.jl a b c</code> . Needs an environment where using <code>GilaElectromagnetics</code> works. Writes the three <code>f_gila_*</code> tables. (a) 37 min, (b) 35 min, (c) 5 s.
<code>refcache.txt</code>	1155 lines: 320-bit <code>pairMom</code> values for the 15 sub-codes at the 11 shapes for $m \in \{-1, 1, 3, 5, 7, 9, 11\}$. Built in 36.7 minutes; read by part (a) so that no cached case is recomputed.
<code>tables/a_correctness.txt</code>	15 sub-codes $\times$ 11 shapes $\times m = -1..12$ against the reference. Global worst 7.35×10^{-44} , 0 of 2310 above 10^{-30} .
<code>tables/b_faces.txt</code>	All 144 face pairs against <code>pairMom</code> at $s = (3/7, 5/11, 2/3)$, $m = -1..8$. 1440 values, worst 1.91×10^{-48} , 0 above 10^{-30} .
<code>tables/c_crosschecks.txt</code>	Against the plain explicit forms; homogeneity; the symmetry sweep; orientation invariance; <code>momentSeries</code> against <code>pairKer</code> ; the <code>tol</code> test.
<code>tables/d_digits.txt</code>	Digits lost in <code>Float64</code> over the 13×14 aspect grid, per sub-code and per m , plus the 15 full grids. Global worst 0.91, 0 cells above 1.0.
<code>tables/e_timing.txt</code>	<code>Float64</code> cost of <code>pairMoments(A,B,12)</code> per sub-code, at the cube and at the slender cell: 87 to 973 μ s.
<code>tables/f_gila.a.txt</code>	<code>wekS/wekE/wekV</code> at <code>intOrd</code> 48 and 64 against the series, 18 entries $\times$ 4 scales $\times$ 2 frequencies, plus the entry-to-geometry map and the $m = -1$ check against Gila's static parts.
<code>tables/f_gila.b.txt</code>	The order-9 fixed rule on all 70 non-touching face pairs of the four offsets, 4 cell scales, against the series and (one representative per sub-code) against <code>pairKer</code> .
<code>tables/f_gila.c.txt</code>	Frequency homogeneity of the series terms at $f = 1, 1 + 0.1i, 0.37$.
<code>moments.tex</code> , <code>moments.pdf</code> <code>figs/</code>	This document. <code>figextra.tex</code> (the grid of the five extra sub-geometries) and <code>figgeo.tex</code> (one macro per sub-geometry, used by the per-geometry figures); the shared TikZ macros and the two original figure grids are input from <code>notes/gen/</code> .

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September 2026 (`notes/integrals.tex`). Referred to throughout as the companion document.

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